TEST DATE AND CONDITIONS		
Date	14/06/2023	
Atmospheric Pressure	999	mB
Ambient Temperature	20.0	°C
Environics Serial No.	5089	

GFM430 Final Inspection & Calibration Check Certificate

GAS DATA LTD
Unit 4, Fairfield Court
Seven Stars Estate
Wheler Rd
Coventry
CV3 4LJ
Tel 02476303311 Fax 02476307711



Customer	Mason Evans
Certificate Number	124289
Order Number	334257

Serial Number	10309
Software Version	G430-00.0024/0013

Recalibration DUE Date
14/06/24

Instrument Checks					
Keyboard	✓		Display Contrast	✓	
Pump Flow In	400	*Accept > 200 cc/min*	Pump Flow @ -200mB	300	*Accept > 200 cc/min*
Clock Set / Running	✓		Labels Fitted	✓	

Gas Checks						
	CH_4		CO_2		O_2	
	Instrument Gas Readings %	True Gas Value %	Instrument Gas Readings %	True Gas Value %	Instrument Gas Readings %	True Gas Value %
Sensor	60.2	60	40.1	40	20.9	20.9
	Accept ±3.0		*Accept ±3.0*		*Accept ±0.5*	
	5.2	5	4.9	5	6.0	6
	Accept ±0.3		*Accept ±0.3*		*Accept ±0.3*	
Zero Reading 100% N2	0.0	0	0.0	0	0.0	0
	Accept ±0.0		*Accept ±0.0*		*Accept ±0.1*	

Optional Gas Checks						
Applied Gas & Range		Concentration Tested @ (ppm)	Instrument Readings *(ppm)*			
			Zero Reading		Instrument Gas Reading	
Toxic Gas	Range *(ppm)*					
H2S	2000	*1500*	0	*Accept ±0.0*	1500	*Accept ±5.0%*
CO	2000	*1000*	0	*Accept ±0.0*	998	*Accept ±5.0%*

TEST DATE AND CONDITIONS		
Date	14-6-23	
Atmospheric Pressure	999	mB
Ambient Temperature	20.0	°C
Environics Serial No.	5089	

GAS DATA LTD
Unit 4
Fairfield Court
Seven Stars Estate
Coventry
CV3 4LJ
UK
+44 (0)24 7630 3311

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GFM430 -1 OUTWARD INSPECTION & QUALITY CHECK SHEET

INSTRUMENT DETAILS			
SO Number	Instrument Type	Instrument Serial Number + SW Version	Job Number(s)
334257	GFM430	10309 G430-24/13	124289

Calibration Technician [redacted] **Date** 14-6-23

Inspection Technician [redacted] **Date** 15.6.23

INSTRUMENT CHECKS			Pass (P), Fail (F) or not applicable (NA)
Function Tests	Dust Caps Fitted		P
	Keyboard Test (All Keys)		P
	Backlight		P
	Clock Set / Running		P
	Comms Test		P
	Pump Flow Test (In & Out)		P
	Overall Leak Test (30mB)		n/a
	Battery Charge Test		P
	Service Date set to?		14.6.23
Channel Test	Data Logging Enabled?		P
	Verify CH4/LEL		P
	Verify CO2		P
	Verify O2		P
	Verify LEL		P
	Verify 1st Option Gas	H2S	P
	Verify 2nd Option Gas	CO	P
	Verify 3rd Option Gas		N/A
	Verify 4th Option Gas		N/A
	Verify Atmospheric pressure		P
	Verify static pressure		N/A
	Verify differential pressure		P
	Verify flow		P
	Verify temperature probe input		P
	Verify vane anemometer input		P
DataBase Checks	Jobcard(s) completed and signed		P
	Jobcard(s) booked off database		P
	Calibration certificate completed		P
	Complete & print QI record		n/a
Label Checks	No. of Calibration label fitted	GDC	11930
	Warranty label fitted		
H2S Range	H2S Range from Sales Order	2000	ppm
	H2S Range from Cal Cert	2000	ppm
	Over-range value correct?		P

INSTRUMENT PACKING LIST		Tick if included
Instrument		✓
Leather Case		X
Instrument Strap		✓
AC Battery Charger (UK)		✓
AC Battery Charger (EURO)		X
AC Battery Charger (US)		X
AC Battery Charger (AUS)		X
Hard Carry Case		✓
Gas Sample Tube – *(new issue)*		✓
Flow Sample Tube – *(new issue)*		✓
Spares Pot		✓
Allen Key		✓
Temperature Probe		X
Vane Anemometer		X
USB Cable		✓
USB Memory stick		X
SiteMan Software	Ver 4.15	X
Internal Filter Pack	Qty	X
External Filter Pack	Qty	X
Field Guide		X
Operation Manual (hard copy)		X

Extra Items:
CAR CHARGER
SAMPLE TUBE

Comments:

TEST DATE AND CONDITIONS		
Date	28/02/2023	
Atmospheric Pressure	1017	mB
Ambient Temperature	19.8	°C
Environics Serial No.	5089	

GAS DATA LTD
Unit 4, Fairfield Court
Seven Stars Estate
Wheler Rd
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Tel 02476303311 Fax 02476307711

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GFM436 Final Inspection & Calibration Check Certificate

Customer	Mason Evans
Certificate Number	123970
Order Number	333729

Serial Number	12984
Software Version	G436-00.0027/0010

Recalibration DUE Date
28/02/24

Instrument Checks					
Keyboard	✓		Display Contrast	✓	
Pump Flow In	500	*Accept > 200 cc/min*	Pump Flow @ -200mB	300	*Accept > 200 cc/min*
Clock Set / Running	✓		Labels Fitted	✓	

Gas Checks						
Sensor	CH_4		CO_2		O_2	
	Instrument Gas Readings %	True Gas Value %	Instrument Gas Readings %	True Gas Value %	Instrument Gas Readings %	True Gas Value %
	59.5	60	39.8	40	20.9	20.9
	Accept ±3.0		*Accept ±3.0*		*Accept ±0.5*	
	4.9	5	4.9	5	6.0	6
	Accept ±0.3		*Accept ±0.3*		*Accept ±0.3*	
Zero Reading 100% N2	0.0	0	0.0	0	0.0	0
	Accept ±0.0		*Accept ±0.0*		*Accept ±0.1*	

Optional Gas Checks						
Applied Gas & Range		Concentration Tested @ (ppm)	Instrument Readings *(ppm)*			
Gas Type	Range *(ppm)*		Zero Reading		Instrument Gas Reading	
H2S	5000	*1500*	0	*Accept ±0.0*	1500	*Accept ±5.0*
CO	2000	*1000*	0	*Accept ±0.0*	998	*Accept ±5.0*
Hexane	*2.0%*	2.0%	0	*Accept ±0.0*	1.99	*Accept ±10.0*

TEST DATE AND CONDITIONS		
Date	28.2.23	
Atmospheric Pressure	1017	mB
Ambient Temperature	19.2	°C
Environics Serial No.	5089	

GAS DATA LTD
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+44 (0)24 7630 3311

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GFM436-1 OUTWARD INSPECTION & QUALITY CHECK SHEET

INSTRUMENT DETAILS			
SO Number	Instrument Type	Instrument Serial Number + SW Version	Job Number(s)
353729	GFM436	[redacted] 27/10	123970

Calibration Technician ..... **Date** 28.2.23

Inspection Technician ..... **Date** 1.3.23

INSTRUMENT CHECKS		Pass (P), Fail (F) or not applicable (NA)	INSTRUMENT PACKING LIST		Tick if included
Function Tests	Dust Caps Fitted	P	Instrument		✓
	Keyboard Test (All Keys)	P	Leather Case		X
	Backlight	P	Instrument Strap		✓
	Clock Set / Running	P	AC Battery Charger (UK)		X
	Comms Test	P	AC Battery Charger (EURO)		X
	Pump Flow Test (In & Out)	P	AC Battery Charger (US)		X
	Overall Leak Test (30mB)	n/a	AC Battery Charger (AUS)		X
	Battery Charge Test	P	Gas Sample Pipe - *(new issue)*		✓
	Service Date set to?	28.2.24	Flow Sample Pipe - *(new issue)*		✓
Channel Tests	Data Logging Enabled?	P	Hard Carry Case		✓
	Verify CH4/LEL/Hexane/PID	P	Spares Pot		✓
	Verify CO2	P	Allen Key		X
	Verify O2	P	Temperature Probe		X
	Verify H2S	P	Vane Anemometer		X
	Verify CO	P	USB Cable		X
	Verify LEL	P	USB Memory stick		✓
	Verify 1st Option Gas	N/A	SM V5 Software	Ver 6.05	✓
	Verify Atmospheric pressure	P	Internal Filter Pack	Qty	X
	Verify differential pressure	P	External Filter Pack	Qty	X
	Verify flow	P	Field Guide		X
	Verify temperature probe input	P	**Extra Items:**		
	Verify vane anemometer input	P			
DataBase Checks	Jobcard(s) completed and signed	P			
	Jobcard(s) booked off database	P			
	Calibration certificate completed	P			
	Complete & print QI record	n/a			
Label Checks	No. of Calibration label fitted	GDC 11778	**Comments:**		
	MCERTS label displayed	N/A			
	Warranty label fitted	P			
H2S Range	H2S Range from Sales Order	5000 ppm			
	H2S Range from Cal Cert	5000 ppm			
	Over-range value correct?	P			

Appendix 8

Chemical Laboratory Analysis Results

Certificate of Analysis

Certificate Number	23-09602	*Issued:*	02-May-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference	23-09602
Client Reference	P23-075
Order No	Aisling Lee
Contract Title	(P23-075) Causewayend, Edinburgh
Description	9 Soil samples, 6 Leachate samples.
Date Received	24-Apr-23
Date Started	24-Apr-23
Date Completed	02-May-23
Test Procedures	Identified by prefix DETSn (details on request).

Notes Opinions and interpretations are outside the laboratory's scope of ISO 17025 accreditation. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced except in full, without the prior written approval of the laboratory.

Approved [redacted]
General Manager

2139

Summary of Chemical Analysis
Soil Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

				Lab No	2160197	2160198	2160199	2160200	2160201	2160202	2160203
				.Sample ID	BH01	BH01	BH01	BH02	BH02	BH02	BH03
				Depth	0.25	1.00	3.00	0.50	2.00	3.00	0.50
				Other ID							
				Sample Type	SOIL	SOIL	SOIL	SOIL	SOIL	SOIL	SOIL
				Sampling Date	17/04/2023	17/04/2023	17/04/2023	18/04/2023	18/04/2023	18/04/2023	19/04/2023
				Sampling Time	n/s	n/s	n/s	n/s	n/s	n/s	n/s
Test	Method	LOD	Units								
Asbestos Quantification	DETSC 1102	0.001	%					0.002			0.010
Preparation											
Moisture Content	DETSC 1004	0.1	%	5.9	14		13	15	13	14	
Metals											
Arsenic	DETSC 2301#	0.2	mg/kg	2.8	7.5		5.2	5.3	5.6	7.1	
Boron, Water Soluble (2.5:1)	DETSC 2311#	0.2	mg/kg	0.4	0.4		1.1	0.5	0.5	1.1	
Cadmium	DETSC 2301#	0.1	mg/kg	0.1	0.2		< 0.1	< 0.1	0.1	0.2	
Chromium	DETSC 2301#	0.15	mg/kg	12	21		21	26	26	25	
Chromium, Hexavalent	DETSC 2204*	1	mg/kg	< 1.0	< 1.0		< 1.0	< 1.0	< 1.0	< 1.0	
Copper	DETSC 2301#	0.2	mg/kg	56	21		22	63	26	31	
Lead	DETSC 2301#	0.3	mg/kg	26	16		23	14	14	40	
Mercury	DETSC 2325#	0.05	mg/kg	0.19	0.35		0.49	0.16	0.70	0.79	
Nickel	DETSC 2301#	1	mg/kg	25	27		26	32	34	31	
Selenium	DETSC 2301#	0.5	mg/kg	< 0.5	< 0.5		< 0.5	< 0.5	< 0.5	< 0.5	
Zinc	DETSC 2301#	1	mg/kg	73	58		51	63	66	94	
Inorganics											
pH	DETSC 2008#		pH	8.2	8.1	8.4	9.0	8.6	8.5	9.7	
Cyanide, Total	DETSC 2130#	0.1	mg/kg	0.2	0.1		0.1	< 0.1	< 0.1	< 0.1	
Total Organic Carbon	DETSC 2084#	0.5	%	0.9	0.8		1.0	1.1	1.1	1.8	
Organic Matter (by calculation)	*	0.1	%	1.5	1.4		1.7	2.0	1.8	3.1	
Sulphate Aqueous Extract as SO4 (2:1)	DETSC 2076#	10	mg/l			23					
Sulphide	DETSC 2024*	10	mg/kg	< 10	< 10		< 10	< 10	< 10	36	
Sulphate as SO4, Total	DETSC 2321#	0.01	%	0.03	0.02		0.06	0.03	0.02	0.11	
Petroleum Hydrocarbons											
Aliphatic C5-C6	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01		< 0.01	< 0.01	< 0.01	< 0.01	
Aliphatic C6-C8	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01		< 0.01	< 0.01	< 0.01	< 0.01	
Aliphatic C8-C10	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01		< 0.01	< 0.01	< 0.01	< 0.01	
Aliphatic C10-C12	DETSC 3072#	1.5	mg/kg	< 1.5	< 1.5		< 1.5	< 1.5	< 1.5	< 1.5	
Aliphatic C12-C16	DETSC 3072#	1.2	mg/kg	< 1.2	< 1.2		< 1.2	< 1.2	< 1.2	12	
Aliphatic C16-C21	DETSC 3072#	1.5	mg/kg	< 1.5	< 1.5		< 1.5	< 1.5	< 1.5	6.5	
Aliphatic C16-C35	DETSC 3072#	4.9	mg/kg	65	< 4.9		< 4.9	< 4.9	< 4.9	6.5	
Aliphatic C21-C35	DETSC 3072#	3.4	mg/kg	65	< 3.4		< 3.4	< 3.4	< 3.4	< 3.4	
Aliphatic C5-C35	DETSC 3072*	10	mg/kg	65	< 10		< 10	< 10	< 10	19	
Aromatic C5-C7	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01		< 0.01	< 0.01	< 0.01	< 0.01	
Aromatic C7-C8	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01		< 0.01	< 0.01	< 0.01	< 0.01	
Aromatic C8-C10	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01		< 0.01	< 0.01	< 0.01	< 0.01	
Aromatic C10-C12	DETSC 3072#	0.9	mg/kg	< 0.9	< 0.9		< 0.9	< 0.9	< 0.9	< 0.9	
Aromatic C12-C16	DETSC 3072#	0.5	mg/kg	< 0.5	< 0.5		< 0.5	< 0.5	< 0.5	< 0.5	
Aromatic C16-C21	DETSC 3072#	0.6	mg/kg	2.2	< 0.6		< 0.6	< 0.6	< 0.6	< 0.6	
Aromatic C21-C35	DETSC 3072#	1.4	mg/kg	130	< 1.4		< 1.4	< 1.4	< 1.4	< 1.4	
Aromatic C5-C35	DETSC 3072*	10	mg/kg	130	< 10		< 10	< 10	< 10	< 10	
TPH Ali/Aro Total C5-C35	DETSC 3072*	10	mg/kg	200	< 10		< 10	< 10	< 10	19	

Key: * -not accredited. # -MCERTS (accreditation only applies if report carries the MCERTS logo). n/s -not supplied.

Summary of Chemical Analysis
Soil Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

				Lab No	2160197	2160198	2160199	2160200	2160201	2160202	2160203
				.Sample ID	BH01	BH01	BH01	BH02	BH02	BH02	BH03
				Depth	0.25	1.00	3.00	0.50	2.00	3.00	0.50
				Other ID							
				Sample Type	SOIL	SOIL	SOIL	SOIL	SOIL	SOIL	SOIL
				Sampling Date	17/04/2023	17/04/2023	17/04/2023	18/04/2023	18/04/2023	18/04/2023	19/04/2023
				Sampling Time	n/s	n/s	n/s	n/s	n/s	n/s	n/s
Test	Method	LOD	Units								
PAHs											
Naphthalene	DETSC 3301	0.1	mg/kg	0.1	< 0.1		0.4	< 0.1	< 0.1	0.2
Acenaphthylene	DETSC 3301	0.1	mg/kg	0.2	< 0.1		< 0.1	< 0.1	< 0.1	1.0
Acenaphthene	DETSC 3301	0.1	mg/kg	0.1	< 0.1		0.8	< 0.1	< 0.1	1.5
Fluorene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1		0.7	< 0.1	< 0.1	0.3
Phenanthrene	DETSC 3301	0.1	mg/kg	0.2	< 0.1		2.1	< 0.1	< 0.1	0.6
Anthracene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1		0.6	< 0.1	< 0.1	0.2
Fluoranthene	DETSC 3301	0.1	mg/kg	0.2	< 0.1		1.6	< 0.1	< 0.1	0.7
Pyrene	DETSC 3301	0.1	mg/kg	0.3	< 0.1		2.0	< 0.1	< 0.1	0.8
Benzo(a)anthracene	DETSC 3301	0.1	mg/kg	0.2	< 0.1		1.1	< 0.1	< 0.1	0.5
Chrysene	DETSC 3301	0.1	mg/kg	0.1	< 0.1		0.9	< 0.1	< 0.1	0.4
Benzo(b)fluoranthene	DETSC 3301	0.1	mg/kg	0.2	< 0.1		0.5	< 0.1	< 0.1	0.3
Benzo(k)fluoranthene	DETSC 3301	0.1	mg/kg	0.1	< 0.1		0.3	< 0.1	< 0.1	0.2
Benzo(a)pyrene	DETSC 3301	0.1	mg/kg	0.2	< 0.1		0.9	< 0.1	< 0.1	0.4
Indeno(1,2,3-c,d)pyrene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1		0.6	< 0.1	< 0.1	0.3
Dibenzo(a,h)anthracene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1		0.2	< 0.1	< 0.1	0.1
Benzo(g,h,i)perylene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1		0.4	< 0.1	< 0.1	0.4
PAH 16 Total	DETSC 3301	1.6	mg/kg	2.1	< 1.6		13	< 1.6	< 1.6	7.6
Phenols											
Phenol - Monohydric	DETSC 2130#	0.3	mg/kg	< 0.3	< 0.3		< 0.3	< 0.3	< 0.3	< 0.3

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Summary of Chemical Analysis
Soil Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

			Lab No	2160204	2160205
			.Sample ID	BH03	BH03
			Depth	1.00	2.00
			Other ID		
			Sample Type	SOIL	SOIL
			Sampling Date	19/04/2023	19/04/2023
			Sampling Time	n/s	n/s
Test	Method	LOD	Units		
Asbestos Quantification	DETSC 1102	0.001	%	0.020	
Preparation					
Moisture Content	DETSC 1004	0.1	%	13	13
Metals					
Arsenic	DETSC 2301#	0.2	mg/kg	6.4	5.4
Boron, Water Soluble (2.5:1)	DETSC 2311#	0.2	mg/kg	1.2	0.3
Cadmium	DETSC 2301#	0.1	mg/kg	0.2	0.2
Chromium	DETSC 2301#	0.15	mg/kg	19	12
Chromium, Hexavalent	DETSC 2204*	1	mg/kg	< 1.0	< 1.0
Copper	DETSC 2301#	0.2	mg/kg	27	28
Lead	DETSC 2301#	0.3	mg/kg	34	8.8
Mercury	DETSC 2325#	0.05	mg/kg	0.28	0.07
Nickel	DETSC 2301#	1	mg/kg	24	17
Selenium	DETSC 2301#	0.5	mg/kg	< 0.5	0.5
Zinc	DETSC 2301#	1	mg/kg	78	29
Inorganics					
pH	DETSC 2008#		pH	9.5	8.6
Cyanide, Total	DETSC 2130#	0.1	mg/kg	< 0.1	< 0.1
Total Organic Carbon	DETSC 2084#	0.5	%	1.1	1.0
Organic Matter (by calculation)	*	0.1	%	1.9	1.7
Sulphate Aqueous Extract as SO4 (2:1)	DETSC 2076#	10	mg/l		
Sulphide	DETSC 2024*	10	mg/kg	28	24
Sulphate as SO4, Total	DETSC 2321#	0.01	%	0.12	0.07
Petroleum Hydrocarbons					
Aliphatic C5-C6	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01
Aliphatic C6-C8	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01
Aliphatic C8-C10	DETSC 3321*	0.01	mg/kg	< 0.01	0.11
Aliphatic C10-C12	DETSC 3072#	1.5	mg/kg	< 1.5	< 1.5
Aliphatic C12-C16	DETSC 3072#	1.2	mg/kg	4.3	< 1.2
Aliphatic C16-C21	DETSC 3072#	1.5	mg/kg	1.8	< 1.5
Aliphatic C16-C35	DETSC 3072#	4.9	mg/kg	< 4.9	< 4.9
Aliphatic C21-C35	DETSC 3072#	3.4	mg/kg	< 3.4	< 3.4
Aliphatic C5-C35	DETSC 3072*	10	mg/kg	< 10	< 10
Aromatic C5-C7	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01
Aromatic C7-C8	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01
Aromatic C8-C10	DETSC 3321*	0.01	mg/kg	< 0.01	0.54
Aromatic C10-C12	DETSC 3072#	0.9	mg/kg	< 0.9	< 0.9
Aromatic C12-C16	DETSC 3072#	0.5	mg/kg	< 0.5	< 0.5
Aromatic C16-C21	DETSC 3072#	0.6	mg/kg	< 0.6	< 0.6
Aromatic C21-C35	DETSC 3072#	1.4	mg/kg	< 1.4	< 1.4
Aromatic C5-C35	DETSC 3072*	10	mg/kg	< 10	< 10
TPH Ali/Aro Total C5-C35	DETSC 3072*	10	mg/kg	< 10	< 10

Key: * -not accredited. # -MCERTS (accreditation only applies if report carries the MCERTS logo). n/s -not supplied.

Summary of Chemical Analysis
Soil Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

			Lab No	2160204	2160205
			.Sample ID	BH03	BH03
			Depth	1.00	2.00
			Other ID		
			Sample Type	SOIL	SOIL
			Sampling Date	19/04/2023	19/04/2023
			Sampling Time	n/s	n/s
Test	Method	LOD	Units		
PAHs					
Naphthalene	DETSC 3301	0.1	mg/kg	0.1	< 0.1
Acenaphthylene	DETSC 3301	0.1	mg/kg	0.8	0.2
Acenaphthene	DETSC 3301	0.1	mg/kg	1.1	0.2
Fluorene	DETSC 3301	0.1	mg/kg	0.2	< 0.1
Phenanthrene	DETSC 3301	0.1	mg/kg	0.4	< 0.1
Anthracene	DETSC 3301	0.1	mg/kg	0.1	< 0.1
Fluoranthene	DETSC 3301	0.1	mg/kg	0.3	< 0.1
Pyrene	DETSC 3301	0.1	mg/kg	0.4	< 0.1
Benzo(a)anthracene	DETSC 3301	0.1	mg/kg	0.3	< 0.1
Chrysene	DETSC 3301	0.1	mg/kg	0.3	< 0.1
Benzo(b)fluoranthene	DETSC 3301	0.1	mg/kg	0.2	< 0.1
Benzo(k)fluoranthene	DETSC 3301	0.1	mg/kg	0.1	< 0.1
Benzo(a)pyrene	DETSC 3301	0.1	mg/kg	0.2	< 0.1
Indeno(1,2,3-c,d)pyrene	DETSC 3301	0.1	mg/kg	0.3	< 0.1
Dibenzo(a,h)anthracene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1
Benzo(g,h,i)perylene	DETSC 3301	0.1	mg/kg	0.2	< 0.1
PAH 16 Total	DETSC 3301	1.6	mg/kg	4.6	< 1.6
Phenols					
Phenol - Monohydric	DETSC 2130#	0.3	mg/kg	< 0.3	< 0.3

Key: * -not accredited. # -MCERTS (accreditation only applies if report carries the MCERTS logo). n/s -not supplied.

Summary of Chemical Analysis
Soil VOC/SVOC Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

Lab No	2160204
.Sample ID	BH03
Depth	1.00
Other ID	
Sample Type	SOIL
Sampling Date	19/04/2023
Sampling Time	n/s

Test	Method	LOD	Units	
VOCs				
Vinyl Chloride	DETSC 3431	0.01	mg/kg	< 0.01
1,1 Dichloroethylene	DETSC 3431	0.01	mg/kg	< 0.01
Trans-1,2-dichloroethylene	DETSC 3431	0.01	mg/kg	< 0.01
1,1-dichloroethane	DETSC 3431	0.01	mg/kg	< 0.01
Cis-1,2-dichloroethylene	DETSC 3431	0.01	mg/kg	< 0.01
2,2-dichloropropane	DETSC 3431	0.01	mg/kg	< 0.01
Bromochloromethane	DETSC 3431	0.01	mg/kg	< 0.01
Chloroform	DETSC 3431	0.01	mg/kg	< 0.01
1,1,1-trichloroethane	DETSC 3431	0.01	mg/kg	< 0.01
1,1-dichloropropene	DETSC 3431	0.01	mg/kg	< 0.01
Carbon tetrachloride	DETSC 3431	0.01	mg/kg	< 0.01
Benzene	DETSC 3431	0.01	mg/kg	< 0.01
1,2-dichloroethane	DETSC 3431	0.01	mg/kg	< 0.01
Trichloroethylene	DETSC 3431	0.01	mg/kg	< 0.01
1,2-dichloropropane	DETSC 3431	0.01	mg/kg	< 0.01
Dibromomethane	DETSC 3431	0.01	mg/kg	< 0.01
Bromodichloromethane	DETSC 3431	0.01	mg/kg	< 0.01
cis-1,3-dichloropropene	DETSC 3431	0.01	mg/kg	< 0.01
Toluene	DETSC 3431	0.01	mg/kg	< 0.01
trans-1,3-dichloropropene	DETSC 3431	0.01	mg/kg	< 0.01
1,1,2-trichloroethane	DETSC 3431	0.01	mg/kg	< 0.01
Tetrachloroethylene	DETSC 3431	0.01	mg/kg	< 0.01
1,3-dichloropropane	DETSC 3431	0.01	mg/kg	< 0.01
Dibromochloromethane	DETSC 3431	0.01	mg/kg	< 0.01
1,2-dibromoethane	DETSC 3431	0.01	mg/kg	< 0.01
Chlorobenzene	DETSC 3431	0.01	mg/kg	< 0.01
1,1,1,2-tetrachloroethane	DETSC 3431	0.01	mg/kg	< 0.01
Ethylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
m+p-Xylene	DETSC 3431	0.01	mg/kg	< 0.01
o-Xylene	DETSC 3431	0.01	mg/kg	< 0.01
Styrene	DETSC 3431*	0.01	mg/kg	< 0.01
Bromoform	DETSC 3431	0.01	mg/kg	< 0.01
Isopropylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
Bromobenzene	DETSC 3431	0.01	mg/kg	< 0.01
1,2,3-trichloropropane	DETSC 3431	0.01	mg/kg	< 0.01
n-propylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
2-chlorotoluene	DETSC 3431	0.01	mg/kg	< 0.01
1,3,5-trimethylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
4-chlorotoluene	DETSC 3431	0.01	mg/kg	< 0.01
Tert-butylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
1,2,4-trimethylbenzene	DETSC 3431	0.01	mg/kg	< 0.01

Key: * -not accredited. n/s -not supplied.

Summary of Chemical Analysis
Soil VOC/SVOC Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

Lab No	2160204
.Sample ID	BH03
Depth	1.00
Other ID	
Sample Type	SOIL
Sampling Date	19/04/2023
Sampling Time	n/s

Test	Method	LOD	Units	
sec-butylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
p-isopropyltoluene	DETSC 3431	0.01	mg/kg	< 0.01
1,3-dichlorobenzene	DETSC 3431	0.01	mg/kg	< 0.01
1,4-dichlorobenzene	DETSC 3431	0.01	mg/kg	< 0.01
n-butylbenzene	DETSC 3431	0.01	mg/kg	< 0.01
1,2-dichlorobenzene	DETSC 3431	0.01	mg/kg	< 0.01
1,2-dibromo-3-chloropropane	DETSC 3431	0.01	mg/kg	< 0.01
1,2,4-trichlorobenzene	DETSC 3431	0.01	mg/kg	< 0.01
Hexachlorobutadiene	DETSC 3431	0.01	mg/kg	< 0.01
1,2,3-trichlorobenzene	DETSC 3431	0.01	mg/kg	< 0.01
MTBE	DETSC 3431*	0.01	mg/kg	< 0.01
SVOCs				
Phenol	DETSC 3433	0.1	mg/kg	< 0.1
Aniline	DETSC 3433*	0.1	mg/kg	< 0.1
2-Chlorophenol	DETSC 3433	0.1	mg/kg	< 0.1
Benzyl Alcohol	DETSC 3433	0.1	mg/kg	< 0.1
2-Methylphenol	DETSC 3433	0.1	mg/kg	< 0.1
Bis(2-chloroisopropyl)ether	DETSC 3433	0.1	mg/kg	< 0.1
3&4-Methylphenol	DETSC 3433	0.1	mg/kg	< 0.1
2,4-Dimethylphenol	DETSC 3433	0.1	mg/kg	< 0.1
Bis-(dichloroethoxy)methane	DETSC 3433	0.1	mg/kg	< 0.1
2,4-Dichlorophenol	DETSC 3433	0.1	mg/kg	< 0.1
1,2,4-Trichlorobenzene	DETSC 3433	0.1	mg/kg	< 0.1
4-Chloro-3-methylphenol	DETSC 3433	0.1	mg/kg	< 0.1
2-Methylnaphthalene	DETSC 3433	0.1	mg/kg	< 0.1
Hexachlorocyclopentadiene	DETSC 3433*	0.1	mg/kg	< 0.1
2,4,6-Trichlorophenol	DETSC 3433	0.1	mg/kg	< 0.1
2,4,5-Trichlorophenol	DETSC 3433*	0.1	mg/kg	< 0.1
2-Chloronaphthalene	DETSC 3433	0.1	mg/kg	< 0.1
2-Nitroaniline	DETSC 3433*	0.1	mg/kg	< 0.1
2,4-Dinitrotoluene	DETSC 3433*	0.1	mg/kg	< 0.1
3-Nitroaniline	DETSC 3433*	0.1	mg/kg	< 0.1
4-Nitrophenol	DETSC 3433*	0.1	mg/kg	< 0.1
Dibenzofuran	DETSC 3433	0.1	mg/kg	< 0.1
2,6-Dinitrotoluene	DETSC 3433	0.1	mg/kg	< 0.1
2,3,4,6-Tetrachlorophenol	DETSC 3433*	0.1	mg/kg	< 0.1
Diethylphthalate	DETSC 3433	0.1	mg/kg	< 0.1
4-Chlorophenylphenylether	DETSC 3433*	0.1	mg/kg	< 0.1
4-Nitroaniline	DETSC 3433*	0.1	mg/kg	< 0.1
2-Methyl-4,6-Dinitrophenol	DETSC 3433*	0.1	mg/kg	< 0.1
Diphenylamine	DETSC 3433	0.1	mg/kg	< 0.1
4-Bromophenylphenylether	DETSC 3433	0.1	mg/kg	< 0.1

Key: * -not accredited. n/s -not supplied.

Summary of Chemical Analysis
Soil VOC/SVOC Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

Lab No	2160204
.Sample ID	BH03
Depth	1.00
Other ID	
Sample Type	SOIL
Sampling Date	19/04/2023
Sampling Time	n/s

Test	Method	LOD	Units	
Hexachlorobenzene	DETSC 3433	0.1	mg/kg	< 0.1
Pentachlorophenol	DETSC 3433*	0.1	mg/kg	< 0.1
Di-n-butylphthalate	DETSC 3433	0.1	mg/kg	< 0.1
Butylbenzylphthalate	DETSC 3433*	0.1	mg/kg	< 0.1
Bis(2-ethylhexyl)phthalate	DETSC 3433	0.1	mg/kg	2.2
Di-n-octylphthalate	DETSC 3433*	0.1	mg/kg	< 0.1
1,4-Dinitrobenzene	DETSC 3433*	0.1	mg/kg	< 0.1
Dimethylphthalate	DETSC 3433	0.1	mg/kg	< 0.1
1,3-Dinitrobenzene	DETSC 3433*	0.1	mg/kg	< 0.1
1,2-Dinitrobenzene	DETSC 3433*	0.1	mg/kg	< 0.1
2,3,5,6-Tetrachlorophenol	DETSC 3433*	0.1	mg/kg	< 0.1
Azobenzene	DETSC 3433	0.1	mg/kg	< 0.1
Carbazole	DETSC 3433*	0.1	mg/kg	< 0.1

WASTE ACCEPTANCE CRITERIA TESTING ANALYTICAL REPORT

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh
Sample Id BH01 1.00

Sample Numbers 2160198 2160206 2160207
Date Analysed 28/04/2023

Test Results On Waste

Determinand and Method Reference	Units	Result	WAC Limit Values: Inert Waste	SNRHW	Hazardous Waste
DETSC 2084# Total Organic Carbon	%	0.8	3	5	6
DETSC 2003# Loss On Ignition	%	2.8	n/a	n/a	10
DETSC 3321# BTEX	mg/kg	< 0.04	6	n/a	n/a
DETSC 3401# PCBs (7 congeners)	mg/kg	< 0.01	1	n/a	n/a
DETSC 3311# TPH (C10 - C40)	mg/kg	< 10	500	n/a	n/a
DETSC 3301 PAHs	mg/kg	< 1.6	100	n/a	n/a
DETSC 2008# pH	pH Units	8.1	n/a	>6	n/a
DETSC 2073* Acid Neutralisation Capacity (pH4)	mol/kg	< 1.0	n/a	TBE	TBE
DETSC 2073* Acid Neutralisation Capacity (pH7)	mol/kg	< 1.0	n/a	TBE	TBE

Test Results On Leachate

Determinand and Method Reference	Conc in Eluate ug/l		Amount Leached* mg/kg		WAC Limit Values Limit values for LS10 Leachate: Inert Waste	SNRHW	Hazardous Waste
	2:1	8:1	LS2	LS10			
DETSC 2306 Arsenic as As	0.21	< 0.16	< 0.002	< 0.01	0.5	2	25
DETSC 2306 Barium as Ba	26	9.8	0.05	0.13	20	100	300
DETSC 2306 Cadmium as Cd	< 0.030	< 0.030	< 0.004	< 0.02	0.04	1	5
DETSC 2306 Chromium as Cr	0.74	< 0.25	< 0.02	< 0.1	0.5	10	70
DETSC 2306 Copper as Cu	2.2	2.1	0.004	0.021	2	50	100
DETSC 2306 Mercury as Hg	< 0.010	< 0.010	< 0.0004	< 0.002	0.01	0.2	2
DETSC 2306 Molybdenum as Mo	< 1.1	< 1.1	< 0.02	< 0.1	0.5	10	30
DETSC 2306 Nickel as Ni	< 0.50	< 0.50	< 0.02	< 0.1	0.4	10	40
DETSC 2306 Lead as Pb	< 0.090	< 0.090	< 0.01	< 0.05	0.5	10	50
DETSC 2306 Antimony as Sb	< 0.17	< 0.17	< 0.01	< 0.05	0.06	0.7	5
DETSC 2306 Selenium as Se	< 0.25	< 0.25	< 0.006	< 0.03	0.1	0.5	7
DETSC 2306 Zinc as Zn	< 1.3	< 1.3	< 0.002	< 0.01	4	50	200
DETSC 2055 Chloride as Cl	5000	2100	< 20	< 100	800	15,000	25,000
DETSC 2055* Fluoride as F	310	< 100	0.62	0.53	10	150	500
DETSC 2055 Sulphate as SO4	9000	2700	< 20	< 100	1000	20,000	50,000
DETSC 2009* Total Dissolved Solids	110000	34000	220	469.3	4000	60,000	100,000
DETSC 2130 Phenol Index	< 100	< 100	< 0.2	< 1	1	n/a	n/a
DETSC 2085 Dissolved Organic Carbon	5200	4300	10.4	< 50	500	800	1000

TBE - To Be Evaluated
SNRHW - Stable Non-Reactive Hazardous Waste

Additional Information

DETSC 2008 pH	7.6	6.4
DETSC 2009 Conductivity uS/cm	157.0	48.3
* Temperature*	17.0	18.0

Mass of Sample Kg*	0.140
Mass of dry Sample Kg*	0.121

Stage 1

Volume of Leachant L2*	0.223
Volume of Eluate VE1*	0.206

Stage 2

Volume of Leachant L8*	0.969
Volume of Eluate VE2*	0.921

Disclaimer: The WAC limit values are provided for guidance only. DETS does not accept responsibility for errors or omissions. Values are correct at time of issue.

V.2.06 * DETS are accredited for the testing of leachates and not the leachate preparation stage which is unaccredited.

WASTE ACCEPTANCE CRITERIA TESTING ANALYTICAL REPORT

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh
Sample Id BH02 2.00

Sample Numbers 2160201 2160208 2160209
Date Analysed 28/04/2023

Test Results On Waste

Determinand and Method Reference	Units	Result	WAC Limit Values: Inert Waste	SNRHW	Hazardous Waste
DETSC 2084# Total Organic Carbon	%	1.1	3	5	6
DETSC 2003# Loss On Ignition	%	3.8	n/a	n/a	10
DETSC 3321# BTEX	mg/kg	< 0.04	6	n/a	n/a
DETSC 3401# PCBs (7 congeners)	mg/kg	< 0.01	1	n/a	n/a
DETSC 3311# TPH (C10 - C40)	mg/kg	27.0	500	n/a	n/a
DETSC 3301 PAHs	mg/kg	< 1.6	100	n/a	n/a
DETSC 2008# pH	pH Units	8.6	n/a	>6	n/a
DETSC 2073* Acid Neutralisation Capacity (pH4)	mol/kg	< 1.0	n/a	TBE	TBE
DETSC 2073* Acid Neutralisation Capacity (pH7)	mol/kg	< 1.0	n/a	TBE	TBE

Test Results On Leachate

Determinand and Method Reference	Conc in Eluate ug/l		Amount Leached* mg/kg		WAC Limit Values Limit values for LS10 Leachate: Inert Waste	SNRHW	Hazardous Waste
	2:1	8:1	LS2	LS10			
DETSC 2306 Arsenic as As	0.25	0.56	< 0.002	< 0.01	0.5	2	25
DETSC 2306 Barium as Ba	21	9.3	0.04	0.11	20	100	300
DETSC 2306 Cadmium as Cd	< 0.030	< 0.030	< 0.004	< 0.02	0.04	1	5
DETSC 2306 Chromium as Cr	0.69	< 0.25	< 0.02	< 0.1	0.5	10	70
DETSC 2306 Copper as Cu	3.2	4.1	0.006	0.039	2	50	100
DETSC 2306 Mercury as Hg	< 0.010	< 0.010	< 0.0004	< 0.002	0.01	0.2	2
DETSC 2306 Molybdenum as Mo	3.9	< 1.1	< 0.02	< 0.1	0.5	10	30
DETSC 2306 Nickel as Ni	< 0.50	< 0.50	< 0.02	< 0.1	0.4	10	40
DETSC 2306 Lead as Pb	< 0.090	0.14	< 0.01	< 0.05	0.5	10	50
DETSC 2306 Antimony as Sb	0.26	0.2	< 0.01	< 0.05	0.06	0.7	5
DETSC 2306 Selenium as Se	3.3	0.61	0.007	< 0.03	0.1	0.5	7
DETSC 2306 Zinc as Zn	< 1.3	< 1.3	< 0.002	< 0.01	4	50	200
DETSC 2055 Chloride as Cl	7700	1500	< 20	< 100	800	15,000	25,000
DETSC 2055* Fluoride as F	490	110	0.98	1.74	10	150	500
DETSC 2055 Sulphate as SO4	19000	3300	38	< 100	1000	20,000	50,000
DETSC 2009* Total Dissolved Solids	100000	31000	200	426.5	4000	60,000	100,000
DETSC 2130 Phenol Index	< 100	< 100	< 0.2	< 1	1	n/a	n/a
DETSC 2085 Dissolved Organic Carbon	4900	3400	< 10	< 50	500	800	1000

TBE - To Be Evaluated
SNRHW - Stable Non-Reactive Hazardous Waste

Additional Information

DETSC 2008 pH	6.3	6.6
DETSC 2009 Conductivity uS/cm	149.0	44.5
* Temperature*	17.0	17.0

Mass of Sample Kg*	0.140
Mass of dry Sample Kg*	0.120

Stage 1

Volume of Leachant L2*	0.219
Volume of Eluate VE1*	0.202

Stage 2

Volume of Leachant L8*	0.957
Volume of Eluate VE2*	0.907

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WASTE ACCEPTANCE CRITERIA TESTING ANALYTICAL REPORT

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh
Sample Id BH03 2.00

Sample Numbers 2160205 2160210 2160211
Date Analysed 28/04/2023

Test Results On Waste

Determinand and Method Reference	Units	Result	WAC Limit Values: Inert Waste	SNRHW	Hazardous Waste
DETSC 2084# Total Organic Carbon	%	1.0	3	5	6
DETSC 2003# Loss On Ignition	%	0.94	n/a	n/a	10
DETSC 3321# BTEX	mg/kg	0.05	6	n/a	n/a
DETSC 3401# PCBs (7 congeners)	mg/kg	< 0.01	1	n/a	n/a
DETSC 3311# TPH (C10 - C40)	mg/kg	< 10	500	n/a	n/a
DETSC 3301 PAHs	mg/kg	< 1.6	100	n/a	n/a
DETSC 2008# pH	pH Units	8.6	n/a	>6	n/a
DETSC 2073* Acid Neutralisation Capacity (pH4)	mol/kg	< 1.0	n/a	TBE	TBE
DETSC 2073* Acid Neutralisation Capacity (pH7)	mol/kg	< 1.0	n/a	TBE	TBE

Test Results On Leachate

Determinand and Method Reference	Conc in Eluate ug/l		Amount Leached* mg/kg		WAC Limit Values Limit values for LS10 Leachate: Inert Waste	SNRHW	Hazardous Waste
	2:1	8:1	LS2	LS10			
DETSC 2306 Arsenic as As	2	1.8	0.004	0.018	0.5	2	25
DETSC 2306 Barium as Ba	38	31	0.08	0.32	20	100	300
DETSC 2306 Cadmium as Cd	< 0.030	< 0.030	< 0.004	< 0.02	0.04	1	5
DETSC 2306 Chromium as Cr	0.54	< 0.25	< 0.02	< 0.1	0.5	10	70
DETSC 2306 Copper as Cu	2.1	2.2	0.004	0.022	2	50	100
DETSC 2306 Mercury as Hg	< 0.010	< 0.010	< 0.0004	< 0.002	0.01	0.2	2
DETSC 2306 Molybdenum as Mo	3.9	< 1.1	< 0.02	< 0.1	0.5	10	30
DETSC 2306 Nickel as Ni	0.76	< 0.50	< 0.02	< 0.1	0.4	10	40
DETSC 2306 Lead as Pb	0.1	0.15	< 0.01	< 0.05	0.5	10	50
DETSC 2306 Antimony as Sb	0.65	0.37	< 0.01	< 0.05	0.06	0.7	5
DETSC 2306 Selenium as Se	0.82	0.32	< 0.006	< 0.03	0.1	0.5	7
DETSC 2306 Zinc as Zn	1.6	< 1.3	0.003	< 0.01	4	50	200
DETSC 2055 Chloride as Cl	7000	1400	< 20	< 100	800	15,000	25,000
DETSC 2055* Fluoride as F	170	< 100	0.34	0.29	10	150	500
DETSC 2055 Sulphate as SO4	56000	8100	112	163.2	1000	20,000	50,000
DETSC 2009* Total Dissolved Solids	130000	43000	260	579.4	4000	60,000	100,000
DETSC 2130 Phenol Index	< 100	< 100	< 0.2	< 1	1	n/a	n/a
DETSC 2085 Dissolved Organic Carbon	4900	5100	< 10	50.7	500	800	1000

TBE - To Be Evaluated
SNRHW - Stable Non-Reactive Hazardous Waste

Additional Information

DETSC 2008 pH	6.7	7.9
DETSC 2009 Conductivity uS/cm	190.0	61.8
* Temperature*	17.0	17.0

Mass of Sample Kg*	0.140
Mass of dry Sample Kg*	0.122

Stage 1

Volume of Leachant L2*	0.227
Volume of Eluate VE1*	0.21

Stage 2

Volume of Leachant L8*	0.979
Volume of Eluate VE2*	0.93

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Summary of Asbestos Analysis
Soil Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

Lab No	Sample ID	Material Type	Result	Comment*	Analyst
2160197	BH01 0.25	SOIL	NAD	none	Vicky Convery
2160198	BH01 1.00	SOIL	NAD	none	Vicky Convery
2160200	BH02 0.50	SOIL	Chrysotile	Chrysotile present as fibre bundles	Vicky Convery
2160201	BH02 2.00	SOIL	NAD	none	Vicky Convery
2160202	BH02 3.00	SOIL	NAD	none	Vicky Convery
2160203	BH03 0.50	SOIL	Amosite	Amosite present in visible Insulating board debris and as fibre bundles	Vicky Convery
2160204	BH03 1.00	SOIL	Amosite	Amosite present in microscopic Insulating board debris and present as fibre bundles	Vicky Convery
2160205	BH03 2.00	SOIL	NAD	none	Vicky Convery

Crocidolite = Blue Asbestos, Amosite = Brown Asbestos, Chrysotile = White Asbestos. Anthophyllite, Actinolite and Tremolite are other forms of Asbestos. Samples are analysed by DETSC 1101 using polarised light microscopy in accordance with HSG248 and documented in-house methods. NAD = No Asbestos Detected. Where a sample is NAD, the result is based on analysis of at least 2 sub-samples and should be taken to mean 'no asbestos detected in sample'. Key: * -not included in laboratory scope of accreditation.

Summary of Asbestos Quantification Analysis
Soil Samples

Our Ref 23-09602
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

		Lab No	2160200	2160203	2160204
		.Sample ID	BH02	BH03	BH03
		Depth	0.50	0.50	1.00
		Other ID			
		Sample Type			
		Sampling Date	18/04/2023	19/04/2023	19/04/2023
		Sampling Time			
Test	Method	Units			
Total Mass% Asbestos (a+b+c)	DETSC 1102	Mass %	**0.002**	**0.010**	**0.020**
Gravimetric Quantification (a)	DETSC 1102	Mass %	na	0.009	0.018
Detailed Gravimetric Quantification (b)	DETSC 1102	Mass %	0.002	<0.001	0.002
Quantification by PCOM (c)	DETSC 1102	Mass %	na	na	na
Potentially Respirable Fibres (d)	DETSC 1102	Fibres/g	na	na	na
Breakdown of Gravimetric Analysis (a)					
Mass of Sample		g	805.93	857.73	837.57
ACMs present*		type		Insulating Board	Insulating Board
Mass of ACM in sample		g		0.20	0.38
% ACM by mass		%		0.02	0.05
% asbestos in ACM		%		40	40
% asbestos in sample		%		0.009	0.018
Breakdown of Detailed Gravimetric Analysis (b)					
% Amphibole bundles in sample		Mass %	na	<0.001	0.002
% Chrysotile bundles in sample		Mass %	0.002	na	na
Breakdown of PCOM Analysis (c)					
% Amphibole fibres in sample		Mass %	na	na	na
% Chrysotile fibres in sample		Mass %	na	na	na
Breakdown of Potentially Respirable Fibre Analysis (d)					
Amphibole fibres		Fibres/g	na	na	na
Chrysotile fibres		Fibres/g	na	na	na

* Denotes test or material description outside of UKAS accreditation.
% asbestos in Asbestos Containing Materials (ACMs) is determined by
by reference to HSG 264.
Recommended sample size for quantification is approximately 1kg
denotes deviating sample

Information in Support of the Analytical Results

Our Ref 23-09602
Client Ref P23-075
Contract (P23-075) Causewayend, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2160197	BH01 0.25 SOIL	17/04/23	GJ 250ml, PT 1L		
2160198	BH01 1.00 SOIL	17/04/23	GJ 250ml, PT 1L		
2160199	BH01 3.00 SOIL	17/04/23	GJ 250ml, PT 1L		
2160200	BH02 0.50 SOIL	18/04/23	GJ 250ml, PT 1L		
2160201	BH02 2.00 SOIL	18/04/23	GJ 250ml, PT 1L		
2160202	BH02 3.00 SOIL	18/04/23	GJ 250ml, PT 1L		
2160203	BH03 0.50 SOIL	19/04/23	GJ 250ml, PT 1L		
2160204	BH03 1.00 SOIL	19/04/23	GJ 250ml, PT 1L		
2160205	BH03 2.00 SOIL	19/04/23	GJ 250ml, PT 1L		
2160206	BH01 1.00 LEACHATE	17/04/23	GJ 250ml, PT 1L		
2160207	BH01 1.00 LEACHATE	17/04/23	GJ 250ml, PT 1L		
2160208	BH02 2.00 LEACHATE	18/04/23	GJ 250ml, PT 1L		
2160209	BH02 2.00 LEACHATE	18/04/23	GJ 250ml, PT 1L		
2160210	BH03 2.00 LEACHATE	19/04/23	GJ 250ml, PT 1L		
2160211	BH03 2.00 LEACHATE	19/04/23	GJ 250ml, PT 1L		

Key: G-Glass P-Plastic J-Jar T-Tub

DETS cannot be held responsible for the integrity of samples received whereby the laboratory did not undertake the sampling. In this instance samples received may be deviating. Deviating Sample criteria are based on British and International standards and laboratory trials in conjunction with the UKAS note 'Guidance on Deviating Samples'. All samples received are listed above. However, those samples that have additional comments in relation to hold time, inappropriate containers etc are deviating due to the reasons stated. This means that the analysis is accredited where applicable, but results may be compromised due to sample deviations. If no sampled date (soils) or date+time (waters) has been supplied then samples are deviating. However, if you are able to supply a sampled date (and time for waters) this will prevent samples being reported as deviating where specific hold times are not exceeded and where the container supplied is suitable.

Soil Analysis Notes

Inorganic soil analysis was carried out on a dried sample, crushed to pass a 425µm sieve, in accordance with BS1377.

Organic soil analysis was carried out on an 'as received' sample. Organics results are corrected for moisture and expressed on a dry weight basis.

The Loss on Drying, used to express organics analysis on an air dried basis, is carried out at a temperature of 28°C +/-2°C.

Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Certificate of Analysis

Certificate Number 23-09883 *Issued:* 11-May-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference 23-09883

Client Reference P23-075

Order No Aisling Lee

Contract Title (P23-075) Causewayend, Edinburgh

Description 3 Soil samples, 4 Leachate samples.

Date Received 26-Apr-23

Date Started 26-Apr-23

Date Completed 11-May-23

Test Procedures Identified by prefix DETSn (details on request).

Notes Opinions and interpretations are outside the laboratory's scope of ISO 17025 accreditation. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced except in full, without the prior written approval of the laboratory.

Approved By [redacted]

Kirk Bridgewood
General Manager

2139

Summary of Chemical Analysis
Soil Samples

Our Ref 23-09883
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

			Lab No	2161770	2161771	2161772
			.Sample ID	BH5	BH6	BH6
			Depth	1.00	0.25	1.00
			Other ID			
			Sample Type	SOIL	SOIL	SOIL
			Sampling Date	20/04/2023	20/04/2023	20/04/2023
			Sampling Time	n/s	n/s	n/s
Test	Method	LOD	Units			
Preparation						
Moisture Content	DETSC 1004	0.1	%	15	5.1	11
Metals						
Arsenic	DETSC 2301#	0.2	mg/kg	6.9	2.7	6.8
Boron, Water Soluble (2.5:1)	DETSC 2311#	0.2	mg/kg	0.2	0.3	0.5
Cadmium	DETSC 2301#	0.1	mg/kg	0.1	< 0.1	0.1
Chromium	DETSC 2301#	0.15	mg/kg	19	15	20
Chromium, Hexavalent	DETSC 2204*	1	mg/kg	< 1.0	< 1.0	< 1.0
Copper	DETSC 2301#	0.2	mg/kg	23	41	31
Lead	DETSC 2301#	0.3	mg/kg	25	12	38
Mercury	DETSC 2325#	0.05	mg/kg	0.08	< 0.05	0.30
Nickel	DETSC 2301#	1	mg/kg	22	25	26
Selenium	DETSC 2301#	0.5	mg/kg	< 0.5	< 0.5	< 0.5
Zinc	DETSC 2301#	1	mg/kg	70	84	73
Inorganics						
pH	DETSC 2008#		pH	8.1	10.1	10.1
Cyanide, Total	DETSC 2130#	0.1	mg/kg	< 0.1	< 0.1	0.4
Total Organic Carbon	DETSC 2084#	0.5	%	0.9	1.2	0.9
Organic Matter (by calculation)	*	0.1	%	1.5	2.0	1.6
Sulphide	DETSC 2024*	10	mg/kg	12	20	28
Sulphate as SO4, Total	DETSC 2321#	0.01	%	0.03	0.06	0.08
Petroleum Hydrocarbons						
Aliphatic C5-C6	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01	< 0.01
Aliphatic C6-C8	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01	< 0.01
Aliphatic C8-C10	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01	< 0.01
Aliphatic C10-C12	DETSC 3072#	1.5	mg/kg	< 1.5	3.3	< 1.5
Aliphatic C12-C16	DETSC 3072#	1.2	mg/kg	< 1.2	110	< 1.2
Aliphatic C16-C21	DETSC 3072#	1.5	mg/kg	< 1.5	100	< 1.5
Aliphatic C16-C35	DETSC 3072#	4.9	mg/kg	< 4.9	160	22
Aliphatic C21-C35	DETSC 3072#	3.4	mg/kg	< 3.4	55	22
Aliphatic C5-C35	DETSC 3072*	10	mg/kg	< 10	270	22
Aromatic C5-C7	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01	< 0.01
Aromatic C7-C8	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01	< 0.01
Aromatic C8-C10	DETSC 3321*	0.01	mg/kg	< 0.01	< 0.01	< 0.01
Aromatic C10-C12	DETSC 3072#	0.9	mg/kg	< 0.9	< 0.9	< 0.9
Aromatic C12-C16	DETSC 3072#	0.5	mg/kg	< 0.5	55	< 0.5
Aromatic C16-C21	DETSC 3072#	0.6	mg/kg	< 0.6	60	< 0.6
Aromatic C21-C35	DETSC 3072#	1.4	mg/kg	< 1.4	110	37
Aromatic C5-C35	DETSC 3072*	10	mg/kg	< 10	220	37
TPH Ali/Aro Total C5-C35	DETSC 3072*	10	mg/kg	< 10	490	60
PAHs						
Naphthalene	DETSC 3301	0.1	mg/kg	< 0.1	0.6	< 0.1

Key: * -not accredited. # -MCERTS (accreditation only applies if report carries the MCERTS logo). n/s -not supplied.

Summary of Chemical Analysis
Soil Samples

Our Ref 23-09883
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

			Lab No	2161770	2161771	2161772
			.Sample ID	BH5	BH6	BH6
			Depth	1.00	0.25	1.00
			Other ID			
			Sample Type	SOIL	SOIL	SOIL
			Sampling Date	20/04/2023	20/04/2023	20/04/2023
			Sampling Time	n/s	n/s	n/s
Test	Method	LOD	Units			
Acenaphthylene	DETSC 3301	0.1	mg/kg	< 0.1	0.2	0.2
Acenaphthene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.4
Fluorene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.3
Phenanthrene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.8
Anthracene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.3
Fluoranthene	DETSC 3301	0.1	mg/kg	0.1	< 0.1	1.1
Pyrene	DETSC 3301	0.1	mg/kg	0.3	0.3	1.6
Benzo(a)anthracene	DETSC 3301	0.1	mg/kg	< 0.1	0.2	0.6
Chrysene	DETSC 3301	0.1	mg/kg	< 0.1	0.2	0.6
Benzo(b)fluoranthene	DETSC 3301	0.1	mg/kg	< 0.1	0.3	0.5
Benzo(k)fluoranthene	DETSC 3301	0.1	mg/kg	< 0.1	0.4	0.3
Benzo(a)pyrene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.7
Indeno(1,2,3-c,d)pyrene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.5
Dibenzo(a,h)anthracene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.2
Benzo(g,h,i)perylene	DETSC 3301	0.1	mg/kg	< 0.1	< 0.1	0.5
PAH 16 Total	DETSC 3301	1.6	mg/kg	< 1.6	2.2	8.6
Phenols						
Phenol - Monohydric	DETSC 2130#	0.3	mg/kg	< 0.3	< 0.3	< 0.3

WASTE ACCEPTANCE CRITERIA TESTING ANALYTICAL REPORT

Our Ref 23-09883
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh
Sample Id BH6 1.00

Sample Numbers 2161772 2161773 2161774
Date Analysed 09/05/2023

Test Results On Waste

Determinand and Method Reference	Units	Result	WAC Limit Values: Inert Waste	SNRHW	Hazardous Waste
DETSC 2084# Total Organic Carbon	%	0.9	3	5	6
DETSC 2003# Loss On Ignition	%	3.4	n/a	n/a	10
DETSC 3321# BTEX	mg/kg	< 0.04	6	n/a	n/a
DETSC 3401# PCBs (7 congeners)	mg/kg	< 0.01	1	n/a	n/a
DETSC 3311# TPH (C10 - C40)	mg/kg	200.0	500	n/a	n/a
DETSC 3301 PAHs	mg/kg	8.6	100	n/a	n/a
DETSC 2008# pH	pH Units	10.1	n/a	>6	n/a
DETSC 2073* Acid Neutralisation Capacity (pH4)	mol/kg	< 1.0	n/a	TBE	TBE
DETSC 2073* Acid Neutralisation Capacity (pH7)	mol/kg	< 1.0	n/a	TBE	TBE

Test Results On Leachate

Determinand and Method Reference	Conc in Eluate ug/l		Amount Leached* mg/kg		WAC Limit Values Limit values for LS10 Leachate: Inert Waste	SNRHW	Hazardous Waste
	2:1	8:1	LS2	LS10			
DETSC 2306 Arsenic as As	3	1.5	0.006	0.018	0.5	2	25
DETSC 2306 Barium as Ba	16	9.2	0.03	0.1	20	100	300
DETSC 2306 Cadmium as Cd	< 0.030	< 0.030	< 0.004	< 0.02	0.04	1	5
DETSC 2306 Chromium as Cr	1.2	< 0.25	< 0.02	< 0.1	0.5	10	70
DETSC 2306 Copper as Cu	4.5	2.9	0.009	0.032	2	50	100
DETSC 2306 Mercury as Hg	< 0.010	0.012	< 0.0004	< 0.002	0.01	0.2	2
DETSC 2306 Molybdenum as Mo	4.7	< 1.1	< 0.02	< 0.1	0.5	10	30
DETSC 2306 Nickel as Ni	0.81	< 0.50	< 0.02	< 0.1	0.4	10	40
DETSC 2306 Lead as Pb	< 0.090	0.12	< 0.01	< 0.05	0.5	10	50
DETSC 2306 Antimony as Sb	0.85	0.23	< 0.01	< 0.05	0.06	0.7	5
DETSC 2306 Selenium as Se	3.7	0.78	0.007	< 0.03	0.1	0.5	7
DETSC 2306 Zinc as Zn	< 1.3	< 1.3	< 0.002	< 0.01	4	50	200
DETSC 2055 Chloride as Cl	2100	2400	< 20	< 100	800	15,000	25,000
DETSC 2055* Fluoride as F	140	< 100	0.28	0.24	10	150	500
DETSC 2055 Sulphate as SO4	130	160	< 20	< 100	1000	20,000	50,000
DETSC 2009* Total Dissolved Solids	140000	37000	280	549.4	4000	60,000	100,000
DETSC 2130 Phenol Index	< 100	< 100	< 0.2	< 1	1	n/a	n/a
DETSC 2085 Dissolved Organic Carbon	8600	5200	17.2	57.9	500	800	1000

TBE - To Be Evaluated
SNRHW - Stable Non-Reactive Hazardous Waste

Additional Information

DETSC 2008 pH	6.2	7.4
DETSC 2009 Conductivity uS/cm	193.0	53.3
* Temperature*	18.0	18.0

Mass of Sample Kg*	0.140
Mass of dry Sample Kg*	0.125

Stage 1

Volume of Leachant L2*	0.234
Volume of Eluate VE1*	0.217

Stage 2

Volume of Leachant L8*	0.997
Volume of Eluate VE2*	0.948

Disclaimer: The WAC limit values are provided for guidance only. DETS does not accept responsibility for errors or omissions. Values are correct at time of issue.

V.2.06 * DETS are accredited for the testing of leachates and not the leachate preparation stage which is unaccredited.

Summary of Chemical Analysis
Leachate Samples

Our Ref 23-09883
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

			Lab No	2165199	2165200
			.Sample ID	BH5	BH6
			Depth	1.00	1.00
			Other ID		
			Sample Type	LEACHATE	LEACHATE
			Sampling Date	20/04/2023	20/04/2023
			Sampling Time	n/s	n/s
Test	Method	LOD	Units		
Preparation					
Leachate 2:1 250g Non-WAC	DETSC 1009*			Y	Y
Metals					
Arsenic, Dissolved	DETSC 2306	0.16	ug/l	0.57	3.2
Cadmium, Dissolved	DETSC 2306	0.03	ug/l	< 0.03	< 0.03
Calcium, Dissolved	DETSC 2306	0.09	mg/l	3.3	17
Chromium, Dissolved	DETSC 2306	0.25	ug/l	0.56	1.2
Copper, Dissolved	DETSC 2306	0.4	ug/l	3.3	6.0
Lead, Dissolved	DETSC 2306	0.09	ug/l	< 0.09	3.7
Magnesium, Dissolved	DETSC 2306	0.02	mg/l	0.30	0.36
Mercury, Dissolved (Low Level)	DETSC 2324	0.001	ug/l	0.028	0.056
Nickel, Dissolved	DETSC 2306	0.5	ug/l	< 0.5	1.4
Selenium, Dissolved	DETSC 2306	0.25	ug/l	< 0.25	0.68
Zinc, Dissolved	DETSC 2306	1.3	ug/l	1.4	2.2
Inorganics					
Total Hardness as CaCO3	DETSC 2303	0.1	mg/l	9.43	44.5
Sulphate as SO4	DETSC 2055	0.1	mg/l	3.9	18
Sulphide	DETSC 2208	10	ug/l	11	< 10
Phenols					
Total Phenol	DETSC 3451*	0.5	ug/l	< 0.50	< 0.50

Key: * -not accredited. n/s -not supplied.

Summary of Asbestos Analysis
Soil Samples

Our Ref 23-09883
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

Lab No	Sample ID	Material Type	Result	Comment*	Analyst
2161770	BH5 1.00	SOIL	NAD	none	Darryl Fletcher
2161771	BH6 0.25	SOIL	NAD	none	Darryl Fletcher
2161772	BH6 1.00	SOIL	NAD	none	Darryl Fletcher

Crocidolite = Blue Asbestos, Amosite = Brown Asbestos, Chrysotile = White Asbestos. Anthophyllite, Actinolite and Tremolite are other forms of Asbestos. Samples are analysed by DETSC 1101 using polarised light microscopy in accordance with HSG248 and documented in-house methods. NAD = No Asbestos Detected. Where a sample is NAD, the result is based on analysis of at least 2 sub-samples and should be taken to mean 'no asbestos detected in sample'. Key: * - not included in laboratory scope of accreditation.

Information in Support of the Analytical Results

Our Ref 23-09883
Client Ref P23-075
Contract (P23-075) Causewayend, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2161770	BH5 1.00 SOIL	20/04/23	GJ 250ml, GJ 60ml		
2161771	BH6 0.25 SOIL	20/04/23	GJ 250ml, GJ 60ml		
2161772	BH6 1.00 SOIL	20/04/23	GJ 250ml, GJ 60ml		
2161773	BH6 1.00 LEACHATE	20/04/23	GJ 250ml, GJ 60ml		
2161774	BH6 1.00 LEACHATE	20/04/23	GJ 250ml, GJ 60ml		
2165199	BH5 1.00 LEACHATE	20/04/23	GJ 250ml, GJ 60ml		
2165200	BH6 1.00 LEACHATE	20/04/23	GJ 250ml, GJ 60ml		

Key: G-Glass J-Jar

DETS cannot be held responsible for the integrity of samples received whereby the laboratory did not undertake the sampling. In this instance samples received may be deviating. Deviating Sample criteria are based on British and International standards and laboratory trials in conjunction with the UKAS note 'Guidance on Deviating Samples'. All samples received are listed above. However, those samples that have additional comments in relation to hold time, inappropriate containers etc are deviating due to the reasons stated. This means that the analysis is accredited where applicable, but results may be compromised due to sample deviations. If no sampled date (soils) or date+time (waters) has been supplied then samples are deviating. However, if you are able to supply a sampled date (and time for waters) this will prevent samples being reported as deviating where specific hold times are not exceeded and where the container supplied is suitable.

Soil Analysis Notes

Inorganic soil analysis was carried out on a dried sample, crushed to pass a 425µm sieve, in accordance with BS1377.

Organic soil analysis was carried out on an 'as received' sample. Organics results are corrected for moisture and expressed on a dry weight basis.

The Loss on Drying, used to express organics analysis on an air dried basis, is carried out at a temperature of 28°C +/-2°C.

Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Certificate of Analysis

Certificate Number 23-10458 *Issued:* 11-May-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference 23-10458

Client Reference P23-075

Order No Aisling Lee

Contract Title (P23-075) Causewayend, Edinburgh

Description One Leachate sample.

Date Received 24-Apr-23

Date Started 02-May-23

Date Completed 11-May-23

Test Procedures Identified by prefix DETSn (details on request).

Notes Opinions and interpretations are outside the laboratory's scope of ISO 17025 accreditation. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced except in full, without the prior written approval of the laboratory.

Approved By [redacted]
General Manager

2139

Summary of Chemical Analysis
Leachate Samples

Our Ref 23-10458
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

Lab No	2165194
.Sample ID	BH01
Depth	0.25
Other ID	
Sample Type	LEACHATE
Sampling Date	17/04/2023
Sampling Time	n/s

Test	Method	LOD	Units	
Preparation				
Leachate 2:1 250g Non-WAC	DETSC 1009*			Y
Metals				
Arsenic, Dissolved	DETSC 2306	0.16	ug/l	0.92
Cadmium, Dissolved	DETSC 2306	0.03	ug/l	< 0.03
Calcium, Dissolved	DETSC 2306	0.09	mg/l	9.8
Chromium, Dissolved	DETSC 2306	0.25	ug/l	< 0.25
Copper, Dissolved	DETSC 2306	0.4	ug/l	4.0
Lead, Dissolved	DETSC 2306	0.09	ug/l	0.81
Magnesium, Dissolved	DETSC 2306	0.02	mg/l	0.68
Mercury, Dissolved (Low Level)	DETSC 2324	0.001	ug/l	0.010
Nickel, Dissolved	DETSC 2306	0.5	ug/l	< 0.5
Selenium, Dissolved	DETSC 2306	0.25	ug/l	0.33
Zinc, Dissolved	DETSC 2306	1.3	ug/l	< 1.3
Inorganics				
Total Hardness as CaCO3	DETSC 2303	0.1	mg/l	27.3
Sulphate as SO4	DETSC 2055	0.1	mg/l	10
Sulphide	DETSC 2208	10	ug/l	16
Phenols				
Total Phenol	DETSC 3451*	0.5	ug/l	< 0.50

Key: * -not accredited. n/s -not supplied.

Information in Support of the Analytical Results

Our Ref 23-10458
Client Ref P23-075
Contract (P23-075) Causewayend, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2165194	BH01 0.25 LEACHATE	17/04/23	GJ 250ml, PT 1L		
2165195	BH02 0.50 LEACHATE	18/04/23	GJ 250ml, PT 1L		
2165196	BH03 0.50 LEACHATE	19/04/23	GJ 250ml, PT 1L		
2165197	BH03 1.00 LEACHATE	19/04/23	GJ 250ml, PT 1L		

Key: G-Glass P-Plastic J-Jar T-Tub

DETS cannot be held responsible for the integrity of samples received whereby the laboratory did not undertake the sampling. In this instance samples received may be deviating. Deviating Sample criteria are based on British and International standards and laboratory trials in conjunction with the UKAS note 'Guidance on Deviating Samples'. All samples received are listed above. However, those samples that have additional comments in relation to hold time, inappropriate containers etc are deviating due to the reasons stated. This means that the analysis is accredited where applicable, but results may be compromised due to sample deviations. If no sampled date (soils) or date+time (waters) has been supplied then samples are deviating. However, if you are able to supply a sampled date (and time for waters) this will prevent samples being reported as deviating where specific hold times are not exceeded and where the container supplied is suitable.

Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Certificate of Analysis

Certificate Number	23-10581	*Issued:*	12-May-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference	23-10581
Client Reference	P23-075
Order No	Aisling Lee
Contract Title	(P23-075) Causewayend, Edinburgh
Description	4 Leachate samples.
Date Received	24-Apr-23
Date Started	03-May-23
Date Completed	12-May-23
Test Procedures	Identified by prefix DETSn (details on request).

Notes Opinions and interpretations are outside the laboratory's scope of ISO 17025 accreditation. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced except in full, without the prior written approval of the laboratory.

Approved [redacted]

Kirk Bridgewood
General Manager

2139

Summary of Chemical Analysis
Leachate Samples

Our Ref 23-10581
Client Ref P23-075
Contract Title (P23-075) Causewayend, Edinburgh

			Lab No	2165997	2165998	2165999	2166000
			.Sample ID	BH01	BH02	BH02	BH03
			Depth	1.00	2.00	3.00	2.00
			Other ID				
			Sample Type	LEACHATE	LEACHATE	LEACHATE	LEACHATE
			Sampling Date	17/04/2023	18/04/2023	18/04/2023	19/04/2023
			Sampling Time	n/s	n/s	n/s	n/s
Test	Method	LOD	Units				
Preparation							
Leachate 2:1 250g Non-WAC	DETSC 1009*			Y	Y	Y	Y
Metals							
Arsenic, Dissolved	DETSC 2306	0.16	ug/l	0.22	< 0.16	< 0.16	1.7
Cadmium, Dissolved	DETSC 2306	0.03	ug/l	< 0.03	< 0.03	< 0.03	< 0.03
Calcium, Dissolved	DETSC 2306	0.09	mg/l	5.8	5.6	6.7	13
Chromium, Dissolved	DETSC 2306	0.25	ug/l	< 0.25	< 0.25	< 0.25	17
Copper, Dissolved	DETSC 2306	0.4	ug/l	3.0	2.7	2.8	2.1
Lead, Dissolved	DETSC 2306	0.09	ug/l	0.14	< 0.09	< 0.09	< 0.09
Magnesium, Dissolved	DETSC 2306	0.02	mg/l	0.45	0.77	0.62	0.40
Mercury, Dissolved (Low Level)	DETSC 2324	0.001	ug/l	0.0027	0.0036	0.026	0.0033
Nickel, Dissolved	DETSC 2306	0.5	ug/l	< 0.5	< 0.5	< 0.5	< 0.5
Selenium, Dissolved	DETSC 2306	0.25	ug/l	< 0.25	< 0.25	< 0.25	0.29
Zinc, Dissolved	DETSC 2306	1.3	ug/l	< 1.3	< 1.3	< 1.3	< 1.3
Inorganics							
Total Hardness as CaCO3	DETSC 2303	0.1	mg/l	16.2	17.3	19.3	34.5
Sulphate as SO4	DETSC 2055	0.1	mg/l	3.1	10	15	23
Sulphide	DETSC 2208	10	ug/l	13	< 10	12	16
Phenols							
Total Phenol	DETSC 3451*	0.5	ug/l	< 0.50	< 0.50	< 0.50	< 0.50

Key: * -not accredited. n/s -not supplied.

Information in Support of the Analytical Results

Our Ref 23-10581
Client Ref P23-075
Contract (P23-075) Causewayend, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2165997	BH01 1.00 LEACHATE	17/04/23	GJ 250ml, PT 1L		
2165998	BH02 2.00 LEACHATE	18/04/23	GJ 250ml, PT 1L		
2165999	BH02 3.00 LEACHATE	18/04/23	GJ 250ml, PT 1L		
2166000	BH03 2.00 LEACHATE	19/04/23	GJ 250ml, PT 1L		

Key: G-Glass P-Plastic J-Jar T-Tub

DETS cannot be held responsible for the integrity of samples received whereby the laboratory did not undertake the sampling. In this instance samples received may be deviating. Deviating Sample criteria are based on British and International standards and laboratory trials in conjunction with the UKAS note 'Guidance on Deviating Samples'. All samples received are listed above. However, those samples that have additional comments in relation to hold time, inappropriate containers etc are deviating due to the reasons stated. This means that the analysis is accredited where applicable, but results may be compromised due to sample deviations. If no sampled date (soils) or date+time (waters) has been supplied then samples are deviating. However, if you are able to supply a sampled date (and time for waters) this will prevent samples being reported as deviating where specific hold times are not exceeded and where the container supplied is suitable.

Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Certificate of Analysis

Certificate Number	23-14024	*Issued:*	21-Jun-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference	23-14024
Client Reference	P23-075
Order No	Lindsay Reid
Contract Title	(P23-075) Causewayside, Edinburgh
Description	One Water sample.
Date Received	14-Jun-23
Date Started	14-Jun-23
Date Completed	21-Jun-23
Test Procedures	Identified by prefix DETSn (details on request).

Notes Opinions and interpretations are outside the laboratory's scope of ISO 17025 accreditation. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced except in full, without the prior written approval of the laboratory.

Approved By [redacted]
Kirk Bridgewood
General Manager

2139

Summary of Chemical Analysis
Water Samples

Our Ref 23-14024
Client Ref P23-075
Contract Title (P23-075) Causewayside, Edinburgh

Test	Method	LOD	Units	Lab No 2186667	.Sample ID BH01/R01	Depth	Other ID	Sample Type WATER	Sampling Date 12/06/2023	Sampling Time n/s
Metals										
Arsenic, Dissolved	DETSC 2306	0.16	ug/l	0.47						
Boron, Dissolved	DETSC 2306*	12	ug/l	150						
Cadmium, Dissolved	DETSC 2306	0.03	ug/l	< 0.03						
Calcium, Dissolved	DETSC 2306	0.09	mg/l	74						
Chromium, Dissolved	DETSC 2306	0.25	ug/l	0.87						
Copper, Dissolved	DETSC 2306	0.4	ug/l	0.9						
Lead, Dissolved	DETSC 2306	0.09	ug/l	< 0.09						
Magnesium, Dissolved	DETSC 2306	0.02	mg/l	13						
Mercury, Dissolved (Low Level)	DETSC 2324	0.001	ug/l	< 0.0010						
Nickel, Dissolved	DETSC 2306	0.5	ug/l	2.4						
Selenium, Dissolved	DETSC 2306	0.25	ug/l	0.39						
Zinc, Dissolved	DETSC 2306	1.3	ug/l	99						
Inorganics										
pH	DETSC 2008		pH	7.4						
Cyanide, Total Low Level	DETSC 2131	0.1	ug/l	1.7						
Dissolved, Oxygen	DETSC 2048*	0.1	mg/l	6.4						
Dissolved Organic Carbon	DETSC 2085	2	mg/l	4.7						
Total Hardness as CaCO3	DETSC 2303	0.1	mg/l	237						
Sulphate as SO4	DETSC 2055	0.1	mg/l	43						
Sulphide	DETSC 2208	10	ug/l	43						
Petroleum Hydrocarbons										
Aliphatic C5-C6	DETSC 3322	0.1	ug/l	< 0.1						
Aliphatic C6-C8	DETSC 3322	0.1	ug/l	< 0.1						
Aliphatic C8-C10	DETSC 3322	0.1	ug/l	< 0.1						
Aliphatic C10-C12	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C12-C16	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C16-C21	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C21-C35	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C5-C35	DETSC 3072*	10	ug/l	< 10						
Aromatic C5-C7	DETSC 3322	0.1	ug/l	< 0.1						
Aromatic C7-C8	DETSC 3322	0.1	ug/l	< 0.1						
Aromatic C8-C10	DETSC 3322	0.1	ug/l	< 0.1						
Aromatic C10-C12	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C12-C16	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C16-C21	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C21-C35	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C5-C35	DETSC 3072*	10	ug/l	< 10						
TPH Ali/Aro Total C5-C35	DETSC 3072*	10	ug/l	< 10						
PAHs										
Naphthalene	DETSC 3304	0.05	ug/l	< 0.05						
Acenaphthylene	DETSC 3304	0.01	ug/l	< 0.01						

Key: * -not accredited. n/s -not supplied.

Summary of Chemical Analysis
Water Samples

Our Ref 23-14024
Client Ref P23-075
Contract Title (P23-075) Causewayside, Edinburgh

Lab No	2186667
.Sample ID	BH01/R01
Depth	
Other ID	
Sample Type	WATER
Sampling Date	12/06/2023
Sampling Time	n/s

Test	Method	LOD	Units	
Acenaphthene	DETSC 3304	0.01	ug/l	< 0.01
Fluorene	DETSC 3304	0.01	ug/l	< 0.01
Phenanthrene	DETSC 3304	0.01	ug/l	0.03
Anthracene	DETSC 3304	0.01	ug/l	< 0.01
Fluoranthene	DETSC 3304	0.01	ug/l	0.01
Pyrene	DETSC 3304	0.01	ug/l	0.01
Benzo(a)anthracene	DETSC 3304*	0.01	ug/l	< 0.01
Chrysene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(b)fluoranthene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(k)fluoranthene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(a)pyrene	DETSC 3304	0.01	ug/l	< 0.01
Indeno(1,2,3-c,d)pyrene	DETSC 3304	0.01	ug/l	< 0.01
Dibenzo(a,h)anthracene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(g,h,i)perylene	DETSC 3304	0.01	ug/l	< 0.01
PAH Total	DETSC 3304	0.2	ug/l	< 0.20
Phenols				
Total Phenol	DETSC 3451*	0.5	ug/l	< 0.50

Key: * -not accredited. n/s -not supplied.

Information in Support of the Analytical Results

Our Ref 23-14024
Client Ref P23-075
Contract (P23-075) Causewayside, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2186667	BH01/R01 WATER	12/06/23	GB 1L x2, GV, PB 1L		

Key: G-Glass P-Plastic B-Bottle V-Vial

DETS cannot be held responsible for the integrity of samples received whereby the laboratory did not undertake the sampling. In this instance samples received may be deviating. Deviating Sample criteria are based on British and International standards and laboratory trials in conjunction with the UKAS note 'Guidance on Deviating Samples'. All samples received are listed above. However, those samples that have additional comments in relation to hold time, inappropriate containers etc are deviating due to the reasons stated. This means that the analysis is accredited where applicable, but results may be compromised due to sample deviations. If no sampled date (soils) or date+time (waters) has been supplied then samples are deviating. However, if you are able to supply a sampled date (and time for waters) this will prevent samples being reported as deviating where specific hold times are not exceeded and where the container supplied is suitable.

Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Certificate of Analysis

Certificate Number	23-12445	*Issued:*	05-Jun-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference	23-12445
Client Reference	P23-015
Order No	Lindsay Reid
Contract Title	(P23-015) Causewayside, Edinburgh
Description	2 Water samples.
Date Received	26-May-23
Date Started	26-May-23
Date Completed	05-Jun-23
Test Procedures	Identified by prefix DETSn (details on request).

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Approved B [redacted]

Kirk Bridgewood
General Manager

2139

Summary of Chemical Analysis
Water Samples

Our Ref 23-12445
Client Ref P23-015
Contract Title (P23-015) Causewayside, Edinburgh

			Lab No	2177339	2177340
			.Sample ID	BH01/R01	BH02/R02
			Depth		
			Other ID		
			Sample Type	WATER	WATER
			Sampling Date	24/05/2023	24/05/2023
			Sampling Time	n/s	n/s
Test	Method	LOD	Units		
Metals					
Arsenic, Dissolved	DETSC 2306	0.16	ug/l	0.95	0.57
Boron, Dissolved	DETSC 2306*	12	ug/l	130	93
Cadmium, Dissolved	DETSC 2306	0.03	ug/l	< 0.03	< 0.03
Calcium, Dissolved	DETSC 2306	0.09	mg/l	63	45
Chromium, Dissolved	DETSC 2306	0.25	ug/l	0.34	< 0.25
Copper, Dissolved	DETSC 2306	0.4	ug/l	< 0.4	1.1
Lead, Dissolved	DETSC 2306	0.09	ug/l	< 0.09	< 0.09
Magnesium, Dissolved	DETSC 2306	0.02	mg/l	18	14
Mercury, Dissolved (Low Level)	DETSC 2324	0.001	ug/l	0.0038	0.0041
Nickel, Dissolved	DETSC 2306	0.5	ug/l	3.2	1.2
Selenium, Dissolved	DETSC 2306	0.25	ug/l	< 0.25	0.57
Zinc, Dissolved	DETSC 2306	1.3	ug/l	14	2.3
Inorganics					
pH	DETSC 2008		pH	6.8	7.8
Cyanide, Total Low Level	DETSC 2131	0.1	ug/l	0.3	0.3
Dissolved, Oxygen	DETSC 2048*	0.1	mg/l	3.7	2.0
Dissolved Organic Carbon	DETSC 2085	2	mg/l	3.2	3.4
Total Hardness as CaCO3	DETSC 2303	0.1	mg/l	229	170
Sulphate as SO4	DETSC 2055	0.1	mg/l	51	69
Sulphide	DETSC 2208	10	ug/l	19	54
Petroleum Hydrocarbons					
Aliphatic C5-C6	DETSC 3322	0.1	ug/l	< 0.1	< 0.1
Aliphatic C6-C8	DETSC 3322	0.1	ug/l	< 0.1	< 0.1
Aliphatic C8-C10	DETSC 3322	0.1	ug/l	< 0.1	< 0.1
Aliphatic C10-C12	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aliphatic C12-C16	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aliphatic C16-C21	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aliphatic C21-C35	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aliphatic C5-C35	DETSC 3072*	10	ug/l	< 10	< 10
Aromatic C5-C7	DETSC 3322	0.1	ug/l	< 0.1	< 0.1
Aromatic C7-C8	DETSC 3322	0.1	ug/l	< 0.1	< 0.1
Aromatic C8-C10	DETSC 3322	0.1	ug/l	< 0.1	< 0.1
Aromatic C10-C12	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aromatic C12-C16	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aromatic C16-C21	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aromatic C21-C35	DETSC 3072*	1	ug/l	< 1.0	< 1.0
Aromatic C5-C35	DETSC 3072*	10	ug/l	< 10	< 10
TPH Ali/Aro Total C5-C35	DETSC 3072*	10	ug/l	< 10	< 10
PAHs					
Naphthalene	DETSC 3304	0.05	ug/l	0.28	< 0.05
Acenaphthylene	DETSC 3304	0.01	ug/l	< 0.01	< 0.01

Key: * -not accredited. n/s -not supplied.

Summary of Chemical Analysis
Water Samples

Our Ref 23-12445
Client Ref P23-015
Contract Title (P23-015) Causewayside, Edinburgh

			Lab No	2177339	2177340
			.Sample ID	BH01/R01	BH02/R02
			Depth		
			Other ID		
			Sample Type	WATER	WATER
			Sampling Date	24/05/2023	24/05/2023
			Sampling Time	n/s	n/s
Test	Method	LOD	Units		
Acenaphthene	DETSC 3304	0.01	ug/l	0.01	< 0.01
Fluorene	DETSC 3304	0.01	ug/l	0.01	0.02
Phenanthrene	DETSC 3304	0.01	ug/l	0.03	0.11
Anthracene	DETSC 3304	0.01	ug/l	< 0.01	0.01
Fluoranthene	DETSC 3304	0.01	ug/l	0.01	0.07
Pyrene	DETSC 3304	0.01	ug/l	0.02	0.09
Benzo(a)anthracene	DETSC 3304*	0.01	ug/l	< 0.01	0.05
Chrysene	DETSC 3304	0.01	ug/l	< 0.01	0.03
Benzo(b)fluoranthene	DETSC 3304	0.01	ug/l	< 0.01	0.06
Benzo(k)fluoranthene	DETSC 3304	0.01	ug/l	< 0.01	0.02
Benzo(a)pyrene	DETSC 3304	0.01	ug/l	< 0.01	0.04
Indeno(1,2,3-c,d)pyrene	DETSC 3304	0.01	ug/l	< 0.01	0.02
Dibenzo(a,h)anthracene	DETSC 3304	0.01	ug/l	< 0.01	< 0.01
Benzo(g,h,i)perylene	DETSC 3304	0.01	ug/l	< 0.01	0.05
PAH Total	DETSC 3304	0.2	ug/l	0.36	0.56
Phenols					
Total Phenol	DETSC 3451*	0.5	ug/l	< 0.50	< 0.50

Key: * -not accredited. n/s -not supplied.

Information in Support of the Analytical Results

Our Ref 23-12445
Client Ref P23-015
Contract (P23-015) Causewayside, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2177339	BH01/R01 WATER	24/05/23	GB 1L x2, GV, PB 1L		
2177340	BH02/R02 WATER	24/05/23	GB 1L x2, GV, PB 1L		

Key: G-Glass P-Plastic B-Bottle V-Vial

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Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Certificate of Analysis

Certificate Number	23-14024	*Issued:*	21-Jun-23

Client Mason Evans Partnership
95 Morrison Street
Glasgow
G5 8BE

Our Reference	23-14024
Client Reference	P23-075
Order No	Lindsay Reid
Contract Title	(P23-075) Causewayside, Edinburgh
Description	One Water sample.
Date Received	14-Jun-23
Date Started	14-Jun-23
Date Completed	21-Jun-23
Test Procedures	Identified by prefix DETSn (details on request).

Notes Opinions and interpretations are outside the laboratory's scope of ISO 17025 accreditation. This certificate is issued in accordance with the accreditation requirements of the United Kingdom Accreditation Service. The results reported herein relate only to the material supplied to the laboratory. This certificate shall not be reproduced except in full, without the prior written approval of the laboratory.

Approved By [redacted]
Kirk Bridgewood
General Manager

2139

Summary of Chemical Analysis
Water Samples

Our Ref 23-14024
Client Ref P23-075
Contract Title (P23-075) Causewayside, Edinburgh

Test	Method	LOD	Units	Lab No 2186667	.Sample ID BH01/R01	Depth	Other ID	Sample Type WATER	Sampling Date 12/06/2023	Sampling Time n/s
Metals										
Arsenic, Dissolved	DETSC 2306	0.16	ug/l	0.47						
Boron, Dissolved	DETSC 2306*	12	ug/l	150						
Cadmium, Dissolved	DETSC 2306	0.03	ug/l	< 0.03						
Calcium, Dissolved	DETSC 2306	0.09	mg/l	74						
Chromium, Dissolved	DETSC 2306	0.25	ug/l	0.87						
Copper, Dissolved	DETSC 2306	0.4	ug/l	0.9						
Lead, Dissolved	DETSC 2306	0.09	ug/l	< 0.09						
Magnesium, Dissolved	DETSC 2306	0.02	mg/l	13						
Mercury, Dissolved (Low Level)	DETSC 2324	0.001	ug/l	< 0.0010						
Nickel, Dissolved	DETSC 2306	0.5	ug/l	2.4						
Selenium, Dissolved	DETSC 2306	0.25	ug/l	0.39						
Zinc, Dissolved	DETSC 2306	1.3	ug/l	99						
Inorganics										
pH	DETSC 2008		pH	7.4						
Cyanide, Total Low Level	DETSC 2131	0.1	ug/l	1.7						
Dissolved, Oxygen	DETSC 2048*	0.1	mg/l	6.4						
Dissolved Organic Carbon	DETSC 2085	2	mg/l	4.7						
Total Hardness as CaCO3	DETSC 2303	0.1	mg/l	237						
Sulphate as SO4	DETSC 2055	0.1	mg/l	43						
Sulphide	DETSC 2208	10	ug/l	43						
Petroleum Hydrocarbons										
Aliphatic C5-C6	DETSC 3322	0.1	ug/l	< 0.1						
Aliphatic C6-C8	DETSC 3322	0.1	ug/l	< 0.1						
Aliphatic C8-C10	DETSC 3322	0.1	ug/l	< 0.1						
Aliphatic C10-C12	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C12-C16	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C16-C21	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C21-C35	DETSC 3072*	1	ug/l	< 1.0						
Aliphatic C5-C35	DETSC 3072*	10	ug/l	< 10						
Aromatic C5-C7	DETSC 3322	0.1	ug/l	< 0.1						
Aromatic C7-C8	DETSC 3322	0.1	ug/l	< 0.1						
Aromatic C8-C10	DETSC 3322	0.1	ug/l	< 0.1						
Aromatic C10-C12	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C12-C16	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C16-C21	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C21-C35	DETSC 3072*	1	ug/l	< 1.0						
Aromatic C5-C35	DETSC 3072*	10	ug/l	< 10						
TPH Ali/Aro Total C5-C35	DETSC 3072*	10	ug/l	< 10						
PAHs										
Naphthalene	DETSC 3304	0.05	ug/l	< 0.05						
Acenaphthylene	DETSC 3304	0.01	ug/l	< 0.01						

Key: * -not accredited. n/s -not supplied.

Summary of Chemical Analysis
Water Samples

Our Ref 23-14024
Client Ref P23-075
Contract Title (P23-075) Causewayside, Edinburgh

Lab No	2186667
.Sample ID	BH01/R01
Depth	
Other ID	
Sample Type	WATER
Sampling Date	12/06/2023
Sampling Time	n/s

Test	Method	LOD	Units	
Acenaphthene	DETSC 3304	0.01	ug/l	< 0.01
Fluorene	DETSC 3304	0.01	ug/l	< 0.01
Phenanthrene	DETSC 3304	0.01	ug/l	0.03
Anthracene	DETSC 3304	0.01	ug/l	< 0.01
Fluoranthene	DETSC 3304	0.01	ug/l	0.01
Pyrene	DETSC 3304	0.01	ug/l	0.01
Benzo(a)anthracene	DETSC 3304*	0.01	ug/l	< 0.01
Chrysene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(b)fluoranthene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(k)fluoranthene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(a)pyrene	DETSC 3304	0.01	ug/l	< 0.01
Indeno(1,2,3-c,d)pyrene	DETSC 3304	0.01	ug/l	< 0.01
Dibenzo(a,h)anthracene	DETSC 3304	0.01	ug/l	< 0.01
Benzo(g,h,i)perylene	DETSC 3304	0.01	ug/l	< 0.01
PAH Total	DETSC 3304	0.2	ug/l	< 0.20
Phenols				
Total Phenol	DETSC 3451*	0.5	ug/l	< 0.50

Key: * -not accredited. n/s -not supplied.

Information in Support of the Analytical Results

Our Ref 23-14024
Client Ref P23-075
Contract (P23-075) Causewayside, Edinburgh

Containers Received & Deviating Samples

Lab No	Sample ID	Date Sampled	Containers Received	Holding time exceeded for tests	Inappropriate container for tests
2186667	BH01/R01 WATER	12/06/23	GB 1L x2, GV, PB 1L		

Key: G-Glass P-Plastic B-Bottle V-Vial

DETS cannot be held responsible for the integrity of samples received whereby the laboratory did not undertake the sampling. In this instance samples received may be deviating. Deviating Sample criteria are based on British and International standards and laboratory trials in conjunction with the UKAS note 'Guidance on Deviating Samples'. All samples received are listed above. However, those samples that have additional comments in relation to hold time, inappropriate containers etc are deviating due to the reasons stated. This means that the analysis is accredited where applicable, but results may be compromised due to sample deviations. If no sampled date (soils) or date+time (waters) has been supplied then samples are deviating. However, if you are able to supply a sampled date (and time for waters) this will prevent samples being reported as deviating where specific hold times are not exceeded and where the container supplied is suitable.

Disposal

From the issue date of this test certificate, samples will be held for the following times prior to disposal :-

Soils - 1 month, Liquids - 2 weeks, Asbestos (test portion) - 6 months

End of Report

Chemical Analysis Summary Table (Soil Samples)
Causewayside, Edinburgh
P23-075

			Sample ID		BH01	BH01	BH01	BH02	BH02	BH02	BH03	BH03	BH03	BH05	BH06	BH06
			Depth (mbgl)		0.25	1.00	3.00	0.50	2.00	3.00	0.50	1.00	2.00	1.00	0.25	1.00
			Sample Type		MADEGROUND	MADEGROUND	NATURAL	MADEGROUND	NATURAL	NATURAL	MADEGROUND	MADEGROUND	MADEGROUND	NATURAL	MADEGROUND	MADEGROUND
			Sampling Date		17/04/2023	17/04/2023	17/04/2023	18/04/2023	18/04/2023	18/04/2023	19/04/2023	19/04/2023	19/04/2023	20/04/2023	20/04/2023	20/04/2023
Test	**Residential with Plant Uptake 1.0% SOM SGV/GSV/SSV (mg/kg)***	**Source**	**Laboratory Limit of Detection (LoD)**	**Units**												
Preparation																
Moisture Content	N/A	N/A	0.1	%	5.9	14	N/T	13	15	13	14	13	13	15	5.1	11
Metals																
Arsenic	38	LQM/CIEH S4ULs (2015)	0.2	mg/kg	2.8	7.5	N/T	5.2	5.3	5.6	7.1	6.4	5.4	6.9	2.7	6.8
Boron (water soluble (2.5:1))	290	LQM/CIEH S4ULs (2015)	0.2	mg/kg	0.4	0.4	N/T	1.1	0.5	0.5	1.1	1.2	0.3	0.2	0.3	0.5
Cadmium	11	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.1	0.2	N/T	< 0.1	< 0.1	0.1	0.2	0.2	0.2	0.1	< 0.1	0.1
Chromium	910	LQM/CIEH S4ULs (2015)	0.15	mg/kg	12	21	N/T	21	26	26	25	19	12	19	15	20
Hexavalent Chromium	6.0	LQM/CIEH S4ULs (2015)	1.0	mg/kg	< 1.0	< 1.0	N/T	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Copper (toxic)	2,400	LQM/CIEH S4ULs (2015)	0.2	mg/kg	56	21	N/T	22	63	26	31	27	28	23	41	31
Copper (phytotoxic)**	200	BS3882:2015	0.2	mg/kg	56	21	N/T	22	63	26	31	27	28	23	41	31
Lead	200	C4SL (DEFRA SP1010 (2014))	0.3	mg/kg	26	16	N/T	23	14	14	40	34	8.8	25	12	38
Mercury (inorganic)	40	LQM/CIEH S4ULs (2015)	0.05	mg/kg	0.19	0.35	N/T	0.49	0.16	0.7	0.79	0.28	0.07	0.08	< 0.05	0.3
Nickel (toxic)	130	LQM/CIEH S4ULs (2015)	1.0	mg/kg	25	27	N/T	26	32	34	31	24	17	22	25	26
Nickel (phytotoxic)**	110	BS3882:2015	1.0	mg/kg	25	27	N/T	26	32	34	31	24	17	22	25	26
Selenium	250	LQM/CIEH S4ULs (2015)	0.5	mg/kg	< 0.5	< 0.5	N/T	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	0.5	< 0.5	< 0.5	< 0.5
Zinc (toxic)	3,700	LQM/CIEH S4ULs (2015)	1.0	mg/kg	73	58	N/T	51	63	66	94	78	29	70	84	73
Zinc (phytotoxic)**	300	BS3882:2015	1.0	mg/kg	73	58	N/T	51	63	66	94	78	29	70	84	73
Inorganics																
pH	N/A	N/A	N/A	pH	8.2	8.1	8.4	9.0	8.6	8.5	9.7	9.5	8.6	8.1	10.1	10.1
Total Cyanide	34	ATRISK (2015)	0.1	mg/kg	0.2	0.1	N/T	0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.4
Total Organic Carbon	N/A	N/A	0.5	%	0.9	0.8	N/T	1.0	1.1	1.1	1.8	1.1	1.0	0.9	1.2	0.9
Organic Matter (by calculation)	N/A	N/A	0.1	%	1.5	1.4	N/T	1.7	2.0	1.8	3.1	1.9	1.7	1.5	2.0	1.6
Sulphate Aqueous Extract as SO4 (2:1)	N/A	N/A	10	mg/l	N/T	N/T	23	N/T	N/T	N/T	N/T	N/T	N/T	N/T	N/T	N/T
Sulphide	N/A	N/A	10	mg/kg	< 10	< 10	N/T	< 10	< 10	< 10	36	28	24	12	20	28
Total Sulphate as SO4	10,000	ICRCL/SAC	0.01	%	0.03	0.02	N/T	0.06	0.03	0.02	0.11	0.12	0.07	0.03	0.06	0.08
Total Petroleum Hydrocarbons (TPHs)																
Aliphatic C5-C6	42	LQM/CIEH S4ULs (2015)	0.01	mg/kg	< 0.01	< 0.01	N/T	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Aliphatic C6-C8	100	LQM/CIEH S4ULs (2015)	0.01	mg/kg	< 0.01	< 0.01	N/T	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Aliphatic C8-C10	27	LQM/CIEH S4ULs (2015)	0.01	mg/kg	< 0.01	< 0.01	N/T	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.11	< 0.01	< 0.01	< 0.01
Aliphatic C10-C12	130	LQM/CIEH S4ULs (2015)	1.5	mg/kg	< 1.5	< 1.5	N/T	< 1.5	< 1.5	< 1.5	< 1.5	< 1.5	< 1.5	< 1.5	3.3	< 1.5
Aliphatic C12-C16	1,100	LQM/CIEH S4ULs (2015)	1.2	mg/kg	< 1.2	< 1.2	N/T	< 1.2	< 1.2	< 1.2	12	4.3	< 1.2	< 1.2	110	< 1.2
Aliphatic C16-C21	65,000	LQM/CIEH S4ULs (2015)	1.5	mg/kg	< 1.5	< 1.5	N/T	< 1.5	< 1.5	< 1.5	6.5	1.8	< 1.5	< 1.5	100	< 1.5
Aliphatic C16-C35	65,000	LQM/CIEH S4ULs (2015)	4.9	mg/kg	65	< 4.9	N/T	< 4.9	< 4.9	< 4.9	6.5	< 4.9	< 4.9	< 4.9	160	22
Aliphatic C21-C35	65,000	LQM/CIEH S4ULs (2015)	3.4	mg/kg	65	< 3.4	N/T	< 3.4	< 3.4	< 3.4	< 3.4	< 3.4	< 3.4	< 3.4	55	22
Aliphatic C5-C35	N/A	N/A	10	mg/kg	65	< 10	N/T	< 10	< 10	< 10	19	< 10	< 10	< 10	270	22
Aromatic C5-C7	70	LQM/CIEH S4ULs (2015)	0.01	mg/kg	< 0.01	< 0.01	N/T	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Aromatic C7-C8	130	LQM/CIEH S4ULs (2015)	0.01	mg/kg	< 0.01	< 0.01	N/T	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
Aromatic C8-C10	34	LQM/CIEH S4ULs (2015)	0.01	mg/kg	< 0.01	< 0.01	N/T	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.54	< 0.01	< 0.01	< 0.01
Aromatic C10-C12	74	LQM/CIEH S4ULs (2015)	0.9	mg/kg	< 0.9	< 0.9	N/T	< 0.9	< 0.9	< 0.9	< 0.9	< 0.9	< 0.9	< 0.9	< 0.9	< 0.9
Aromatic C12-C16	140	LQM/CIEH S4ULs (2015)	0.5	mg/kg	< 0.5	< 0.5	N/T	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	55	< 0.5
Aromatic C16-C21	260	LQM/CIEH S4ULs (2015)	0.6	mg/kg	2.2	< 0.6	N/T	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	60	< 0.6
Aromatic C21-C35	1,100	LQM/CIEH S4ULs (2015)	1.4	mg/kg	130	< 1.4	N/T	< 1.4	< 1.4	< 1.4	< 1.4	< 1.4	< 1.4	< 1.4	110	37
Aromatic C5-C35	N/A	N/A	10	mg/kg	130	< 10	N/T	< 10	< 10	< 10	< 10	< 10	< 10	< 10	220	37
Total Aliphatics and Aromatics C5-C35	N/A	N/A	10	mg/kg	200	< 10	N/T	< 10	< 10	< 10	19	< 10	< 10	< 10	490	60
Polycyclic Aromatic Hydrocarbons (PAHs)																
Naphthalene	2.3	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.1	< 0.1	N/T	0.4	< 0.1	< 0.1	0.2	0.1	< 0.1	< 0.1	0.6	< 0.1
Acenaphthylene	170	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.2	< 0.1	N/T	< 0.1	< 0.1	< 0.1	1.0	0.8	0.2	< 0.1	0.2	0.2
Acenaphthene	210	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.1	< 0.1	N/T	0.8	< 0.1	< 0.1	1.5	1.1	0.2	< 0.1	< 0.1	0.4
Fluorene	170	LQM/CIEH S4ULs (2015)	0.1	mg/kg	< 0.1	< 0.1	N/T	0.7	< 0.1	< 0.1	0.3	0.2	< 0.1	< 0.1	< 0.1	0.3
Phenanthrene	95	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.2	< 0.1	N/T	2.1	< 0.1	< 0.1	0.6	0.4	< 0.1	< 0.1	< 0.1	0.8
Anthracene	2,400	LQM/CIEH S4ULs (2015)	0.1	mg/kg	< 0.1	< 0.1	N/T	0.6	< 0.1	< 0.1	0.2	0.1	< 0.1	< 0.1	< 0.1	0.3
Fluoranthene	280	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.2	< 0.1	N/T	1.6	< 0.1	< 0.1	0.7	0.3	< 0.1	0.1	< 0.1	1.1
Pyrene	620	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.3	< 0.1	N/T	2.0	< 0.1	< 0.1	0.8	0.4	< 0.1	0.3	0.3	1.6
Benzo(a)anthracene	7.2	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.2	< 0.1	N/T	1.1	< 0.1	< 0.1	0.5	0.3	< 0.1	< 0.1	0.2	0.6
Chrysene	15	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.1	< 0.1	N/T	0.9	< 0.1	< 0.1	0.4	0.3	< 0.1	< 0.1	0.2	0.6
Benzo(b)fluoranthene	2.6	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.2	< 0.1	N/T	0.5	< 0.1	< 0.1	0.3	0.2	< 0.1	< 0.1	0.3	0.5
Benzo(k)fluoranthene	77	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.1	< 0.1	N/T	0.3	< 0.1	< 0.1	0.2	0.1	< 0.1	< 0.1	0.4	0.3
Benzo(a)pyrene	2.2	LQM/CIEH S4ULs (2015)	0.1	mg/kg	0.2	< 0.1	N/T	0.9	< 0.1	< 0.1	0.4	0.2	< 0.1	< 0.1	< 0.1	0.7
Indeno(1,2,3-c,d)pyrene	27	LQM/CIEH S4ULs (2015)	0.1	mg/kg	< 0.1	< 0.1	N/T	0.6	< 0.1	< 0.1	0.3	0.3	< 0.1	< 0.1	< 0.1	0.5
Dibenzo(a,h)anthracene	0.24	LQM/CIEH S4ULs (2015)	0.1	mg/kg	< 0.1	< 0.1	N/T	0.2	< 0.1	< 0.1	0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.2
Benzo(g,h,i)perylene	320	LQM/CIEH S4ULs (2015)	0.1	mg/kg	< 0.1	< 0.1	N/T	0.4	< 0.1	< 0.1	0.4	0.2	< 0.1	< 0.1	< 0.1	0.5
Total PAHs	N/A	N/A	1.6	mg/kg	2.1	< 1.6	N/T	13	< 1.6	< 1.6	7.6	4.6	< 1.6	< 1.6	2.2	8.6
Phenols																
Phenol - Monohydric	120	LQM/CIEH S4ULs (2015)	0.3	mg/kg	< 0.3	< 0.3	N/T	< 0.3	< 0.3	< 0.3	< 0.3	< 0.3	< 0.3	< 0.3	< 0.3	< 0.3

* Average SOM calculated to be 1.836%.

** Based on a pH of>7.0 (average pH calculated to be 8.9)

*** In the absense of a formalised guideline value for lead, we have been conservative and utilised the provisional C4SL guideline value as outlined in the DEFRA SP1010 document.

**** Most conservative laboratory limit of detection (LoD) used as no reporting value was available in the LQM/CIEH s4ULs (2015), ATRISK (2015) or C4SL (DEFRA SP1010, 2014)

+ Based on p-xylene as the most conservative value

Toxic exceedances above relevant assessment criteria

Phytotoxic exceedances above relevant assessment criteria

Above the s4ULs guideline value but below the laboratory limit of detection (LoD)

N/T Not Tested

Chemical Analysis Summary Table (Soil Samples)
Causewayside, Edinburgh
P23-075

			Sample ID		BH01	BH01	BH01	BH02	BH02	BH02	BH03	BH03	BH03	BH05	BH06	BH06
			Depth (mbgl)		0.25	1.00	3.00	0.50	2.00	3.00	0.50	1.00	2.00	1.00	0.25	1.00
			Sample Type		MADEGROUND	MADEGROUND	NATURAL	MADEGRO UND	NATURAL	NATURAL	MADEGROUND	MADEGROUND	MADEGROUND	NATURAL	MADEGRO UND	MADEGROUND
			Sampling Date		17/04/2023	17/04/2023	17/04/2023	18/04/2023	18/04/2023	18/04/2023	19/04/2023	19/04/2023	19/04/2023	20/04/2023	20/04/2023	20/04/2023
Test	**Residential with Plant Uptake 1.0% SOM SGV/GSV/SSV (mg/kg)***	**Source**	**Laboratory Limit of Detection (LoD)**	**Units**												
Volatile Organic Compounds (VOCs)																
Vinyl Chloride	0.00064	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,1 Dichloroethylene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Trans-1,2-dichloroethylene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,1-dichloroethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Cis-1,2-dichloroethylene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
2,2-dichloropropane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Bromochloromethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Chloroform****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,1,1-trichloroethane	8.8	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,1-dichloropropene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Carbon tetrachloride****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Benzene	0.087	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2-dichloroethane	0.0071	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Trichloroethylene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2-dichloropropane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Dibromomethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Bromodichloromethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
cis-1,3-dichloropropene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Toluene	130	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
trans-1,3-dichloropropene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,1,2-trichloroethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Tetrachloroethylene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,3-dichloropropane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Dibromochloromethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2-dibromoethane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Chlorobenzene***	0.5	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,1,1,2-tetrachloroethane	1.2	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Ethylbenzene	47	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
m+p-Xylene+	56	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
o-Xylene+	60	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Styrene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Bromoform****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Isopropylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Bromobenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2,3-trichloropropane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
n-propylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
2-chlorotoluene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,3,5-trimethylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
4-chlorotoluene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Tert-butylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2,4-trimethylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
sec-butylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
p-isopropyltoluene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,3-dichlorobenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,4-dichlorobenzene	61	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
n-butylbenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2-dichlorobenzene	23	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2-dibromo-3-chloropropane****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2,4-trichlorobenzene	2.6	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Hexachlorobutadiene	0.7	LQM/CIEH S4ULs (2015)	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
1,2,3-trichlorobenzene****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
MTBE****	0.01	LoD	0.01	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.01	N/T	N/T	N/T	N/T
Semi-Volatile Organic Compounds (SVOCs)																
Phenol	120	LQM/CIEH S4ULs (2015)	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Aniline****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2-Chlorophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Benzyl Alcohol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2-Methylphenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Bis(2-chloroisopropyl)ether****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
3&4-Methylphenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,4-Dimethylphenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Bis-(dichloroethoxy)methane****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,4-Dichlorophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
1,2,4-Trichlorobenzene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
4-Chloro-3-methylphenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2-Methylnaphthalene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Hexachlorocyclopentadiene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,4,6-Trichlorophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,4,5-Trichlorophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2-Chloronaphthalene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2-Nitroaniline****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,4-Dinitrotoluene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
3-Nitroaniline****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
4-Nitrophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Dibenzofuran****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,6-Dinitrotoluene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,3,4,6-Tetrachlorophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Diethylphthalate****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
4-Chlorophenylphenylether****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
4-Nitroaniline****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2-Methyl-4,6-Dinitrophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Diphenylamine****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
4-Bromophenylphenylether****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Hexachlorobenzene	0.2	LQM/CIEH S4ULs (2015)	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Pentachlorophenol	0.22	LQM/CIEH S4ULs (2015)	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Di-n-butylphthalate****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Butylbenzylphthalate****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Bis(2-ethylhexyl)phthalate****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	2.2	N/T	N/T	N/T	N/T
Di-n-octylphthalate****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
1,4-Dinitrobenzene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Dimethylphthalate****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
1,3-Dinitrobenzene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
1,2-Dinitrobenzene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
2,3,5,6-Tetrachlorophenol****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Azobenzene****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T
Carbazole****	0.1	LoD	0.1	mg/kg	N/T	N/T	N/T	N/T	N/T	N/T	N/T	< 0.1	N/T	N/T	N/T	N/T

* Average SOM calculated to be 1.836%.

** Based on a pH of>7.0 (average pH calculated to be 8.9)

*** In the absense of a formalised guideline value for lead, we have been conservative and utilised the provisional C4SL guideline value as outlined in the DEFRA SP1010 document.

**** Most conservative laboratory limit of detection (LoD) used as no reporting value was available in the LQM/CIEH s4ULs (2015), ATRISK (2015) or C4SL (DEFRA SP1010, 2014)

+ Based on p-xylene as the most conservative value

	Toxic exceedances above relevant assessment criteria
	Phytotoxic exceedances above relevant assessment criteria
	Above the s4ULs guideline value but below the laboratory limit of detection (LoD)
N/T	Not Tested

Chemical Analysis Summary Table (Soil Samples - Asbestos Screen (plus quantification, where applicable))

Causewayside, Edinburgh

P23-075

Sample ID	Depth (m bgl)	Material Type	Result	Comment*	Quantification (%)	Analyst
BH01	0.25	MADEGROUND	NAD	none	N/A	Vicky Convery
BH01	1.00	MADEGROUND	NAD	none	N/A	Vicky Convery
BH02	0.50	MADEGROUND	Chrysotile	Chrysotile present as fibre bundles	0.002	Vicky Convery
BH02	2.00	NATURAL	NAD	none	N/A	Vicky Convery
BH02	3.00	NATURAL	NAD	none	N/A	Vicky Convery
BH03	0.50	MADEGROUND	Amosite	Amosite present in visible insulating board debris and as fibre bundles	0.010	Vicky Convery
BH03	1.00	MADEGROUND	Amosite	Amosite present in microscopic insulating board debris and present as fibre bundles	0.020	Vicky Convery
BH03	2.00	MADEGROUND	NAD	none	N/A	Vicky Convery
BH05	1.00	NATURAL	NAD	none	N/A	Darryl Fletcher
BH06	0.25	MADEGROUND	NAD	none	N/A	Darryl Fletcher
BH06	1.00	MADEGROUND	NAD	none	N/A	Darryl Fletcher

*Crocidolite = Blue Asbestos, Amosite = Brown Asbestos, Chrysotile = White Asbestos. Anthophyllite, Actinolite and Tremolite are other forms of Asbestos. Samples are analysed by DETSC 1101 using polarised light microscopy in accordance with HSG248 and documented in-house methods. NAD = No Asbestos Detected. Where a sample is NAD, the result is based on analysis of at least 2 sub-samples and should be taken to mean 'no asbestos detected in sample'. Key: * -not included in laboratory scope of accreditation.*

NAD - No asbestos detected

Chemical Analysis Summary Table (Leachate Analysis - Surface Water Assessment)

Causewayside, Edinburgh

P23-075

			Sample ID	BH01	BH01	BH02	BH02	BH03	BH05	BH06
			Depth (mbgl)	0.25	1.00	2.00	3.00	2.00	1.00	1.00
			Sample Type	MADEGROUND	MADEGROUND	MADEGROUND	NATURAL	MADEGROUND	MADEGROUND	MADEGROUND
			Sampling Date	17/04/2023	17/04/2023	18/04/2023	18/04/2023	19/04/2023	20/04/2023	20/04/2023
Test	**Relevant Guideline (EQS)**	**Source**	**Laboratory Limit of Detection (LoD)**	**Units**						
Metals										
Arsenic	50	WAT-SG-53 (April 2020)	0.16	ug/l	0.92	0.22	< 0.16	< 0.16	1.7	0.57	3.2
Cadmium	<0.08	WAT-SG-53 (April 2020)	0.03	ug/l	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
Calcium	N/A	N/A	0.09	mg/l	9.8	5.8	5.6	6.7	13	3.3	17
Chromium	4.7	WAT-SG-53 (April 2020)	0.25	ug/l	< 0.25	< 0.25	< 0.25	< 0.25	17	0.56	1.2
Copper	1.0	WAT-SG-53 (April 2020)	0.4	ug/l	4.0	3.0	2.7	2.8	2.1	3.3	6.0
Lead	1.2	WAT-SG-53 (April 2020)	0.09	ug/l	0.81	0.14	< 0.09	< 0.09	< 0.09	< 0.09	3.7
Magnesium	N/A	N/A	0.02	mg/l	0.68	0.45	0.77	0.62	0.4	0.3	0.36
Mercury (inorganic (low level))	0.07	WAT-SG-53 (April 2020)	0.001	ug/l	0.01	0.0027	0.0036	0.026	0.0033	0.028	0.056
Nickel	4.0	WAT-SG-53 (April 2020)	0.5	ug/l	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	1.4
Selenium**	0.25	LoD	0.25	ug/l	0.33	< 0.25	< 0.25	< 0.25	0.29	< 0.25	0.68
Zinc	10.9	WAT-SG-53 (April 2020)	1.3	ug/l	< 1.3	< 1.3	< 1.3	< 1.3	< 1.3	1.4	2.2
Inorganics											
Total Hardness as CaCO3	N/A	N/A	0.1	mg/l	27.3	16.2	17.3	19.3	34.5	9.43	44.5
Sulphate as SO4	400	WAT-SG-53 (April 2020)	0.1	mg/l	10	3.1	10	15	23	3.9	18
Sulphide	N/A	N/A	10	ug/l	16	13	< 10	12	16	11	< 10
Phenols											
Total Phenol**	7.7	LoD	0.5	ug/l	< 0.50	< 0.50	< 0.50	< 0.50	< 0.50	< 0.50	< 0.50

* Based on an average water hardness, from all samples tested, of 24.075 mg/l

** Most conservative laboratory limit of detection (LoD) used as no reporting value was available in the WAT-SG-53 (April 2020) document.

Exceedances above relevant assessment criteria

Chemical Analysis Summary Table (Groundwater Samples - Surface Water Assessment)

Causewayside, Edinburgh

P23-075

					Round 1		Round 2		Round 3
				Sample ID	BH01B/RBH01B	BH02A/RBH02A	BH01B/RBH01B	BH02A/RBH02A	BH01B/RBH01B
				Sample Type	GROUNDWATER	GROUNDWATER	GROUNDWATER	GROUNDWATER	GROUNDWATER
				Sampling Date	10/05/2023	10/05/2023	24/05/2023	24/05/2023	12/06/2023
Test	**Relevant Guideline (EQS)**	**Source**	**Laboratory Limit of Detection (LoD)**	**Units**					
Metals									
Arsenic	50	WAT-SG-53 (April 2020)	0.16	ug/l	0.87	0.72	0.95	0.57	0.47
Boron	2000	WAT-SG-53 (April 2020)	12	ug/l	120	89	130	93	150
Cadmium	0.09	WAT-SG-53 (April 2020)	0.03	ug/l	0.07	< 0.03	< 0.03	< 0.03	< 0.03
Calcium	N/A	N/A	0.09	mg/l	62	33	63	45	74
Chromium	4.7	WAT-SG-53 (April 2020)	0.25	ug/l	0.7	0.89	0.34	< 0.25	0.87
Copper	1.0	WAT-SG-53 (April 2020)	0.4	ug/l	4.6	1.4	< 0.4	1.1	0.9
Lead	1.2	WAT-SG-53 (April 2020)	0.09	ug/l	0.25	< 0.09	< 0.09	< 0.09	< 0.09
Magnesium	N/A	N/A	0.02	mg/l	19	9.7	18	14	13
Mercury (inorganic (low level))	0.07	WAT-SG-53 (April 2020)	0.001	ug/l	0.0054	0.0045	0.0038	0.0041	< 0.0010
Nickel	4.0	WAT-SG-53 (April 2020)	0.5	ug/l	3.2	1.2	3.2	1.2	2.4
Selenium**	0.25	LoD	0.25	ug/l	0.52	2.0	< 0.25	0.57	0.39
Zinc	10.9	WAT-SG-53 (April 2020)	1.3	ug/l	49	32	14	2.3	99
Inorganics									
pH	N/A	N/A	N/A	pH	7.4	7.7	6.8	7.8	7.4
Total Cyanide (low level)	1.0	WAT-SG-53 (April 2020)	0.1	ug/l	0.3	0.4	0.3	0.3	1.7
Dissolved Oxygen	N/A	N/A	0.1	mg/l	2.1	3.0	3.7	2.0	6.4
Dissolved Organic Carbon	N/A	N/A	2	mg/l	3.2	3.6	3.2	3.4	4.7
Total Hardness as CaCO3	N/A	n/A	0.1	mg/l	233	121	229	170	237
Sulphate as SO4	400	WAT-SG-53 (April 2020)	0.1	mg/l	51	52	51	69	43
Sulphide	N/A	N/A	10	ug/l	37	17	19	54	43
Total Petroleum Hydrocarbons (TPHs)									
Aliphatic C5-C6**	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aliphatic C6-C8**	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aliphatic C8-C10**	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aliphatic C10-C12**	1.0	LoD	1.0	ug/l	5.7	< 1.0	< 1.0	< 1.0	< 1.0
Aliphatic C12-C16**	1.0	LoD	1.0	ug/l	27	4.9	< 1.0	< 1.0	< 1.0
Aliphatic C16-C21**	1.0	LoD	1.0	ug/l	33	42	< 1.0	< 1.0	< 1.0
Aliphatic C21-C35**	1.0	LoD	1.0	ug/l	400	230	< 1.0	< 1.0	< 1.0
Aliphatic C5-C35	N/A	N/A	10	ug/l	460	280	< 10	< 10	< 10
Aromatic C5-C7**	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aromatic C7-C8**	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aromatic C8-C10**	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aromatic C10-C12**	1.0	LoD	1.0	ug/l	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C12-C16**	1.0	LoD	1.0	ug/l	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C16-C21**	1.0	LoD	1.0	ug/l	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C21-C35**	1.0	LoD	1.0	ug/l	59	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C5-C35	N/A	N/A	10	ug/l	59	< 10	< 10	< 10	< 10
Total Aliphatics and Aromatics C5-C35	N/A	N/A	10	ug/l	520	280	< 10	< 10	< 10
Polycyclic Aromatic Hydrocarbons (PAHs)									
Naphthalene	2.0	WAT-SG-53 (April 2020)	0.05	ug/l	< 0.50	< 0.50	0.28	< 0.05	< 0.05
Acenaphthylene**	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	< 0.01	< 0.01
Acenaphthene**	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	0.01	< 0.01	< 0.01
Fluorene**	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	0.01	0.02	< 0.01
Phenanthrene**	0.01	LoD	0.01/0.10	ug/l	0.22	0.48	0.03	0.11	0.03
Anthracene	0.1	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.01	< 0.01
Fluoranthene	0.0063	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	0.3	0.01	0.07	0.01
Pyrene**	0.01/0.10	LoD	0.01/0.10	ug/l	0.14	0.29	0.02	0.09	0.01
Benzo(a)anthracene**	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	0.14	< 0.01	0.05	< 0.01
Chrysene**	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	0.11	< 0.01	0.03	< 0.01
Benzo(b)fluoranthene	0.00017	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	0.2	< 0.01	0.06	< 0.01
Benzo(k)fluoranthene	0.00017	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.02	< 0.01
Benzo(a)pyrene	0.00017	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.04	< 0.01
Indeno(1,2,3-c,d)pyrene	0.00017	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.02	< 0.01
Dibenzo(a,h)anthracene**	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	< 0.01	< 0.01
Benzo(g,h,i)perylene	0.00017	WAT-SG-53 (April 2020)	0.01/0.10	ug/l	< 0.10	0.12	< 0.01	0.05	< 0.01
Total PAHs	N/A	N/A	0.2	ug/l	< 2.00	< 2.00	0.36	0.56	< 0.20
Phenols									
Total Phenol	7.7	WAT-SG-53 (April 2020)	0.5/5.0	ug/l	< 5.00	< 5.00	< 0.50	< 0.50	< 0.50

* Based on an average water hardness from all groundwater obtained of 198 mg/l

** Most conservative laboratory limit of detection (LoD) used as no reporting value was available in the WAT-SG-53 (April 2020) document.

Exceedances above relevant assessment criteria

Laboratory limit of detection (LoD) above current guideline value

Chemical Analysis Summary Table (Leachate Analysis - Groundwater Assessment)

Causewayside, Edinburgh

P23-075

			Sample ID		BH01	BH01	BH02	BH02	BH03	BH05	BH06
			Depth (mbgl)		0.25	1.00	2.00	3.00	2.00	1.00	1.00
			Sample Type		MADEGROUND	MADEGROUND	MADEGROUND	NATURAL	MADEGROUND	MADEGROUND	MADEGROUND
			Sampling Date		17/04/2023	17/04/2023	18/04/2023	18/04/2023	19/04/2023	20/04/2023	20/04/2023
Test	**Relevant Guideline**	**Source**	**Laboratory Limit of Detection (LoD)**	**Units**							
Metals											
Arsenic	10	WAT-PS-01-01 (August 2014)	0.16	ug/l	0.92	0.22	< 0.16	< 0.16	1.7	0.57	3.2
Cadmium	0.1	WAT-PS-01-01 (August 2014)	0.03	ug/l	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
Calcium	N/A	N/A	0.09	mg/l	9.8	5.8	5.6	6.7	13	3.3	17
Chromium	50	WAT-PS-01-01 (August 2014)	0.25	ug/l	< 0.25	< 0.25	< 0.25	< 0.25	17	0.56	1.2
Copper	2000	WHO (4th ed. 2011)	0.4	ug/l	4.0	3.0	2.7	2.8	2.1	3.3	6.0
Lead	10	DWI PR14 Guidance	0.09	ug/l	0.81	0.14	< 0.09	< 0.09	< 0.09	< 0.09	3.7
Magnesium	N/A	N/A	0.02	mg/l	0.68	0.45	0.77	0.62	0.4	0.3	0.36
Mercury (inorganic (low level))	0.01	WAT-PS-01-01 (August 2014)	0.001	ug/l	0.01	0.0027	0.0036	0.026	0.0033	0.028	0.056
Nickel	20	WAT-PS-01-01 (August 2014)	0.5	ug/l	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	1.4
Selenium	10	WAT-PS-01-01 (August 2014)	0.25	ug/l	0.33	< 0.25	< 0.25	< 0.25	0.29	< 0.25	0.68
Zinc	3000	WHO (4th ed. 2011)	1.3	ug/l	< 1.3	< 1.3	< 1.3	< 1.3	< 1.3	1.4	2.2
Inorganics											
Total Hardness as CaCO3	N/A	N/A	0.1	mg/l	27.3	16.2	17.3	19.3	34.5	9.43	44.5
Sulphate as SO4	250	WAT-PS-01-01 (August 2014)	0.1	mg/l	10	3.1	10	15	23	3.9	18
Sulphide	N/A	N/A	10	ug/l	16	13	< 10	12	16	11	< 10
Phenols											
Total Phenol*	0.5	LoD	0.5	ug/l	< 0.50	< 0.50	< 0.50	< 0.50	< 0.50	< 0.50	< 0.50

* Most conservative laboratory limit of detection (LoD) used as no reporting value was available in the WAT-PS-10-01 (August 2014) or WHO Document (2011). Note that the WHO guideline values have only been utilised for metals.

Exceedances above relevant assessment criteria

Chemical Analysis Summary Table (Groundwater Samples - Groundwater Assessment)
Causewayside, Edinburgh
P23-075

					Round 1		Round 2		Round 3
			Sample ID		BH01B/RBH01B	BH02A/RBH02A	BH01B/RBH01B	BH02A/RBH02A	BH01B/RBH01B
			Sample Type		GROUNDWATER	GROUNDWATER	GROUNDWATER	GROUNDWATER	GROUNDWATER
			Sampling Date		10/05/2023	10/05/2023	24/05/2023	24/05/2023	12/06/2023
Test	**Relevant Guideline**	**Source**	**Laboratory Limit of Detection (LoD)**	**Units**					
Metals									
Arsenic	10	WAT-PS-10-01 (August 2014)	0.16	ug/l	0.87	0.72	0.95	0.57	0.47
Boron	1000	WAT-PS-10-01 (August 2014)	12	ug/l	120	89	130	93	150
Cadmium	0.1	WAT-PS-10-01 (August 2014)	0.03	ug/l	0.07	< 0.03	< 0.03	< 0.03	< 0.03
Calcium	N/A	N/A	0.09	mg/l	62	33	63	45	74
Chromium	50	WAT-PS-10-01 (August 2014)	0.25	ug/l	0.7	0.89	0.34	< 0.25	0.87
Copper	2000	WHO (4th ed. 2011)	0.4	ug/l	4.6	1.4	< 0.4	1.1	0.9
Lead	10	DWI PR14 Guidance	0.09	ug/l	0.25	< 0.09	< 0.09	< 0.09	< 0.09
Magnesium	N/A	N/A	0.02	mg/l	19	9.7	18	14	13
Mercury (inorganic (low level))	0.01	WAT-PS-10-01 (August 2014)	0.001	ug/l	0.0054	0.0045	0.0038	0.0041	< 0.0010
Nickel	20	WAT-PS-10-01 (August 2014)	0.5	ug/l	3.2	1.2	3.2	1.2	2.4
Selenium	10	WAT-PS-10-01 (August 2014)	0.25	ug/l	0.52	2.0	< 0.25	0.57	0.39
Zinc	3000	WHO (4th ed. 2011)	1.3	ug/l	49	32	14	2.3	99
Inorganics									
pH	N/A	N/A	N/A	pH	7.4	7.7	6.8	7.8	7.4
Total Cyanide (low level)	50	WAT-PS-10-01 (August 2014)	0.1	ug/l	0.3	0.4	0.3	0.3	1.7
Dissolved Oxygen	N/A	N/A	0.1	mg/l	2.1	3.0	3.7	2.0	6.4
Dissolved Organic Carbon	N/A	N/A	2	mg/l	3.2	3.6	3.2	3.4	4.7
Total Hardness as CaCO3	N/A	N/A	0.1	mg/l	233	121	229	170	237
Sulphate as SO4	250	WAT-PS-10-01 (August 2014)	0.1	mg/l	51	52	51	69	43
Sulphide	N/A	N/A	10	ug/l	37	17	19	54	43
Total Petroleum Hydrocarbons (TPHs)									
Aliphatic C5-C6*	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aliphatic C6-C8*	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aliphatic C8-C10*	0.1	LoD	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aliphatic C10-C12*	1.0	LoD	1.0	ug/l	5.7	< 1.0	< 1.0	< 1.0	< 1.0
Aliphatic C12-C16*	1.0	LoD	1.0	ug/l	27	4.9	< 1.0	< 1.0	< 1.0
Aliphatic C16-C21*	1.0	LoD	1.0	ug/l	33	42	< 1.0	< 1.0	< 1.0
Aliphatic C21-C35*	1.0	LoD	1.0	ug/l	400	230	< 1.0	< 1.0	< 1.0
Aliphatic C5-C35	N/A	N/A	10	ug/l	460	280	< 10	< 10	< 10
Aromatic C5-C7+	1.0	WAT-PS-10-01 (August 2014)	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aromatic C7-C8+	300	WAT-PS-10-01 (August 2014)	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aromatic C8-C10++	4.0	WAT-PS-10-01 (August 2014)	0.1	ug/l	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Aromatic C10-C12*	1.0	LoD	1.0	ug/l	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C12-C16*	1.0	LoD	1.0	ug/l	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C16-C21*	1.0	LoD	1.0	ug/l	< 1.0	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C21-C35*	1.0	LoD	1.0	ug/l	59	< 1.0	< 1.0	< 1.0	< 1.0
Aromatic C5-C35	N/A	N/A	10	ug/l	59	< 10	< 10	< 10	< 10
Total Aliphatics and Aromatics C5-C35	N/A	N/A	10	ug/l	520	280	< 10	< 10	< 10
Polycyclic Aromatic Hydrocarbons (PAHs)									
Naphthalene*	0.05	LoD	0.05	ug/l	< 0.50	< 0.50	0.28	< 0.05	< 0.05
Acenaphthylene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	< 0.01	< 0.01
Acenaphthene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	0.01	< 0.01	< 0.01
Fluorene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	0.01	0.02	< 0.01
Phenanthrene*	0.01/0.10	LoD	0.01/0.10	ug/l	0.22	0.48	0.03	0.11	0.03
Anthracene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.01	< 0.01
Fluoranthene	0.1	WAT-PS-10-01 (August 2014)	0.01/0.10	ug/l	< 0.10	0.3	0.01	0.07	0.01
Pyrene*	0.01/0.10	LoD	0.01/0.10	ug/l	0.14	0.29	0.02	0.09	0.01
Benzo(a)anthracene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	0.14	< 0.01	0.05	< 0.01
Chrysene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	0.11	< 0.01	0.03	< 0.01
Benzo(b)fluoranthene	0.1	WAT-PS-10-01 (August 2014)	0.01/0.10	ug/l	< 0.10	0.2	< 0.01	0.06	< 0.01
Benzo(k)fluoranthene	0.1	WAT-PS-10-01 (August 2014)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.02	< 0.01
Benzo(a)pyrene	0.01	WAT-PS-10-01 (August 2014)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.04	< 0.01
Indeno(1,2,3-c,d)pyrene	0.1	WAT-PS-10-01 (August 2014)	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	0.02	< 0.01
Dibenzo(a,h)anthracene*	0.01/0.10	LoD	0.01/0.10	ug/l	< 0.10	< 0.10	< 0.01	< 0.01	< 0.01
Benzo(g,h,i)perylene	0.1	WAT-PS-10-01 (August 2014)	0.01/0.10	ug/l	< 0.10	0.12	< 0.01	0.05	< 0.01
Total PAHs	N/A	N/A	0.2	ug/l	< 2.00	< 2.00	0.36	0.56	< 0.20
Phenols									
Total Phenol	0.5/5.0	LoD	0.5/5.0	ug/l	< 5.00	< 5.00	< 0.50	< 0.50	< 0.50

* Most conservative laboratory limit of detection (LoD) used as no reporting value was available in the WAT-PS-10-01 (August 2014) or WHO Document (2011). Note that the WHO guideline values have only been utilised for metals.

** Guideline value for Xylene utlilised

+ Guideline values of 1.0 ug/l (benzene) and 4.0 ug/l (toluene) have been developed within this aromatic fraction range - the most conservative guideline value has been utilised.

++ Guideline values of 300 ug/l (ethylbenzene) and 3.0 ug/l (xylenes) have been developed within this aromatic fraction range - the most conservative guideline value has been utilised.

Exceedances above relevant assessment criteria

Laboratory limit of detection (LoD) above current guideline value

Appendix 9

RTM Groundwater Monitoring Worksheets

Surface Water Assessment

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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The calculation of equations in this worksheet has been independently checked by Entec (UK) Ltd on behalf of the Environment Agency.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C10-C12			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu) | Variable | Value | Unit | Source

Contaminant		Aliphatic C10-C12		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	5.70E-03	mg/l	*Max conc recorded within BH01B/RBH01B*
Half life for degradation of contaminant in water	$t_{1/2}$	1.02E+02	days	*Howard et al*
Calculated decay rate		6.80E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH01B/RBH01B*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.14E+04	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	3.16E+05	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.14E+04	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	m

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

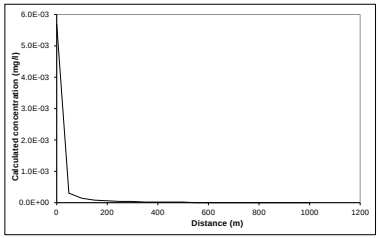
Calculated Parameters | Variable

Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	5.51E+04	fraction
Decay rate used		1.23E-07	d^{-1}
Rate of contaminant flow due to retardation	u	5.39E-05	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	2.49E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.29E+03**	

Remedial Targets

Remedial Target		**2.29E+00**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**2.49E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	5.7E-03
50.0	2.98E-04
100.0	1.38E-04
150.0	8.38E-05
200.0	5.73E-05
250.0	4.17E-05
300.0	3.16E-05
350.0	2.46E-05
400.0	1.96E-05
450.0	1.58E-05
500.0	1.30E-05
550.0	1.07E-05
600.0	8.91E-06
650.0	7.48E-06
700.0	6.31E-06
750.0	5.35E-06
800.0	4.56E-06
850.0	3.90E-06
900.0	3.35E-06
950.0	2.88E-06
1000.0	2.49E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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The calculation of equations in this worksheet has been independently checked by Entec (UK) Ltd on behalf of the Environment Agency.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C12-C16			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu) | Variable | Value | Unit | Source

Contaminant		Aliphatic C12-C16		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	6.98E-03	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	4.40E+02	days	*Howard et al*
Calculated decay rate		1.58E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.81E+05	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters | Variable

Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	8.74E+05	fraction
Decay rate used		1.80E-09	d^{-1}
Rate of contaminant flow due to retardation	u	3.40E-06	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	2.50E-05	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.79E+02**	

Remedial Targets

Remedial Target		**2.79E-01**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**2.50E-05**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)

Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	5.01E+06	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.81E+05	l/kg

Define dispersivity (click brown cell and use pull down list)

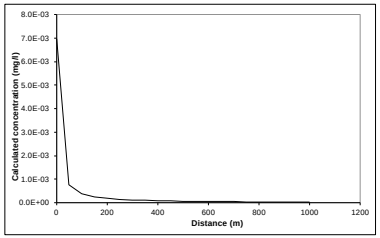
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	7.0E-03
50.0	7.55E-04
100.0	3.81E-04
150.0	2.51E-04
200.0	1.84E-04
250.0	1.44E-04
300.0	1.18E-04
350.0	9.84E-05
400.0	8.41E-05
450.0	7.29E-05
500.0	6.40E-05
550.0	5.68E-05
600.0	5.08E-05
650.0	4.58E-05
700.0	4.14E-05
750.0	3.77E-05
800.0	3.45E-05
850.0	3.17E-05
900.0	2.92E-05
950.0	2.70E-05
1000.0	2.50E-05

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C16-C21			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Aliphatic C16-C21		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.56E-02	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	5.30E+02	days	*Howard et al*
Calculated decay rate		1.31E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.44E+07	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	6.94E+07	fraction
Decay rate used		1.88E-11	d^{-1}
Rate of contaminant flow due to retardation	u	4.28E-08	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	6.06E-05	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.57E+02**	

Remedial Targets

Remedial Target		**2.57E-01**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**6.06E-05**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	3.98E+08	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.44E+07	l/kg

Define dispersivity (click brown cell and use pull down list)

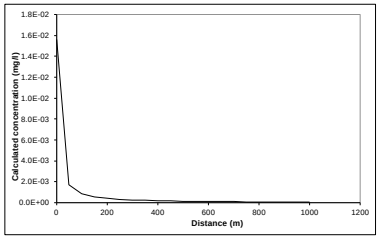
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.6E-02
50.0	1.70E-03
100.0	8.59E-04
150.0	5.67E-04
200.0	4.19E-04
250.0	3.29E-04
300.0	2.69E-04
350.0	2.26E-04
400.0	1.94E-04
450.0	1.69E-04
500.0	1.49E-04
550.0	1.33E-04
600.0	1.19E-04
650.0	1.08E-04
700.0	9.81E-05
750.0	8.97E-05
800.0	8.24E-05
850.0	7.59E-05
900.0	7.02E-05
950.0	6.51E-05
1000.0	6.06E-05

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C21-C35			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
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The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Aliphatic C21-C35		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.27E-01	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	7.30E+02	days	*Howard et al*
Calculated decay rate		9.50E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.44E+07	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	6.94E+07	fraction
Decay rate used		1.37E-11	d^{-1}
Rate of contaminant flow due to retardation	u	4.28E-08	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	5.50E-04	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.30E+02**	

Remedial Targets

Remedial Target		**2.30E-01**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**5.50E-04**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	3.98E+08	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.44E+07	l/kg

Define dispersivity (click brown cell and use pull down list)

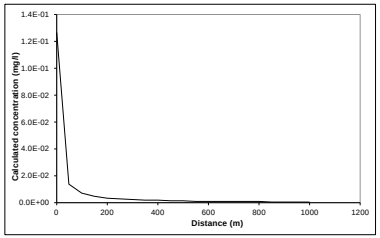
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.3E-01
50.0	1.38E-02
100.0	7.05E-03
150.0	4.68E-03
200.0	3.48E-03
250.0	2.75E-03
300.0	2.26E-03
350.0	1.91E-03
400.0	1.65E-03
450.0	1.44E-03
500.0	1.28E-03
550.0	1.15E-03
600.0	1.04E-03
650.0	9.42E-04
700.0	8.61E-04
750.0	7.92E-04
800.0	7.31E-04
850.0	6.78E-04
900.0	6.30E-04
950.0	5.88E-04
1000.0	5.50E-04

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aromatic C21-C35			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Aromatic C21-C35		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	5.90E-02	mg/l	*Max conc recorded within BH01B/RBH01B*
Half life for degradation of contaminant in water	$t_{1/2}$	7.30E+02	days	*Howard et al*
Calculated decay rate		9.50E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH01B/RBH01B*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	4.56E+03	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	2.20E+04	fraction
Decay rate used		4.32E-08	d^{-1}
Rate of contaminant flow due to retardation	u	1.35E-04	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.28E-04	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**4.59E+02**	

Remedial Targets

Remedial Target		**4.59E-01**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.28E-04**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.26E+05	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	4.56E+03	l/kg

Define dispersivity (click brown cell and use pull down list)

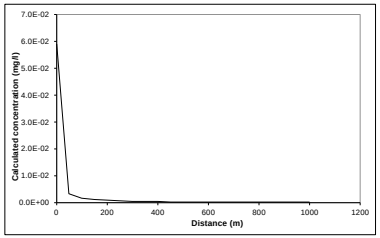
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	5.9E-02
50.0	3.35E-03
100.0	1.67E-03
150.0	1.10E-03
200.0	8.18E-04
250.0	6.45E-04
300.0	5.30E-04
350.0	4.48E-04
400.0	3.86E-04
450.0	3.38E-04
500.0	2.99E-04
550.0	2.68E-04
600.0	2.42E-04
650.0	2.20E-04
700.0	2.01E-04
750.0	1.85E-04
800.0	1.71E-04
850.0	1.58E-04
900.0	1.47E-04
950.0	1.37E-04
1000.0	1.28E-04

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Benzo(a)anthracene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Benzo(a)anthracene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.40E-04	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.36E+03	days	*Howard et al*
Calculated decay rate		5.09E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	5.36E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	2.59E+02	fraction
Decay rate used		1.96E-06	d^{-1}
Rate of contaminant flow due to retardation	u	1.15E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	2.16E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**6.48E+02**	

Remedial Targets

Remedial Target		**6.48E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**2.16E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.48E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	5.36E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

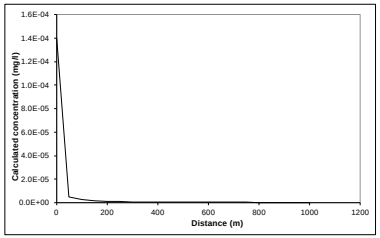
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.4E-04
50.0	4.97E-06
100.0	2.49E-06
150.0	1.65E-06
200.0	1.23E-06
250.0	9.77E-07
300.0	8.08E-07
350.0	6.87E-07
400.0	5.97E-07
450.0	5.26E-07
500.0	4.69E-07
550.0	4.23E-07
600.0	3.85E-07
650.0	3.52E-07
700.0	3.24E-07
750.0	3.00E-07
800.0	2.79E-07
850.0	2.61E-07
900.0	2.44E-07
950.0	2.29E-07
1000.0	2.16E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Benzo(a)pyrene			
Target Concentration (C_T)	0.00000017	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Benzo(a)pyrene		from Level 1
Target Concentration	C_T	1.70E-07	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	4.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.06E+03	days	*Howard et al*
Calculated decay rate		6.55E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.46E+02	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	4.02E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.46E+02	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

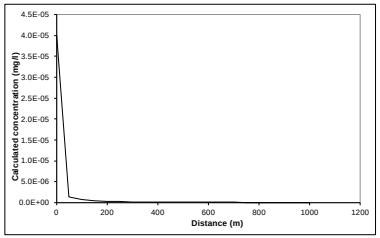
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	7.03E+02	fraction
Decay rate used		9.32E-07	d^{-1}
Rate of contaminant flow due to retardation	u	4.23E-03	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	5.89E-08	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**6.79E+02**	

Remedial Targets

Remedial Target		**1.16E-04**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**5.89E-08**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	4.0E-05
50.0	1.42E-06
100.0	7.08E-07
150.0	4.69E-07
200.0	3.48E-07
250.0	2.76E-07
300.0	2.28E-07
350.0	1.93E-07
400.0	1.67E-07
450.0	1.47E-07
500.0	1.31E-07
550.0	1.18E-07
600.0	1.07E-07
650.0	9.76E-08
700.0	8.97E-08
750.0	8.28E-08
800.0	7.68E-08
850.0	7.15E-08
900.0	6.68E-08
950.0	6.26E-08
1000.0	5.89E-08

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Benzo(b)fluoranthene			
Target Concentration (C_T)	0.00000017	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Benzo(b)fluoranthene		from Level 1
Target Concentration	C_T	1.70E-07	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	3.80E-05	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	1.22E+03	days	*Howard et al*
Calculated decay rate		5.69E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.60E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	7.80E+01	fraction
Decay rate used		7.29E-06	d^{-1}
Rate of contaminant flow due to retardation	u	3.81E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.87E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.04E+02**	

Remedial Targets

Remedial Target		**3.46E-05**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.87E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)

Soil water partition coefficient Kd — l/kg

Entry for non-polar organic chemicals (option)

Fraction of organic carbon in aquifer foc: 3.62E-02 fraction

Organic carbon partition coefficient Koc: 4.41E+02 l/kg

Entry for ionic organic chemicals (option)

Sorption coefficient for related species $K_{oc,n}$ — l/kg

Sorption coefficient for ionised species $K_{oc,i}$ — l/kg

pH value pH

acid dissociation constant pKa

Fraction of organic carbon in aquifer foc — fraction

Soil water partition coefficient Kd: 1.60E+01 l/kg

Define dispersivity (click brown cell and use pull down list)

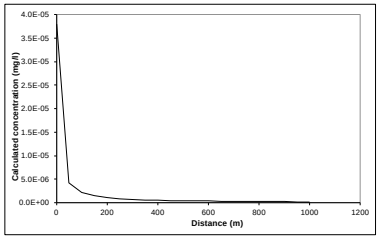
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	3.8E-05
50.0	4.18E-06
100.0	2.14E-06
150.0	1.43E-06
200.0	1.07E-06
250.0	8.50E-07
300.0	7.04E-07
350.0	5.98E-07
400.0	5.19E-07
450.0	4.58E-07
500.0	4.08E-07
550.0	3.68E-07
600.0	3.34E-07
650.0	3.06E-07
700.0	2.82E-07
750.0	2.60E-07
800.0	2.42E-07
850.0	2.26E-07
900.0	2.11E-07
950.0	1.98E-07
1000.0	1.87E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Benzo(ghi)perylene			
Target Concentration (C_T)	0.00000017	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
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Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Benzo(ghi)perylene		from Level 1
Target Concentration	C_T	1.70E-07	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	5.80E-05	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	1.31E+03	days	*Howard et al*
Calculated decay rate		5.28E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.89E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	5.23E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.89E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

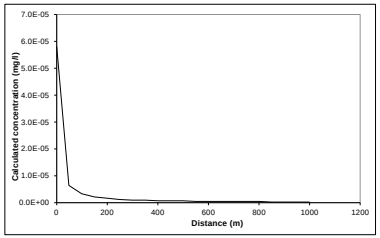
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	9.22E+01	fraction
Decay rate used		5.72E-06	d^{-1}
Rate of contaminant flow due to retardation	u	3.22E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	2.89E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.01E+02**	

Remedial Targets

Remedial Target		**3.42E-05**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**2.89E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	5.8E-05
50.0	6.38E-06
100.0	3.27E-06
150.0	2.19E-06
200.0	1.64E-06
250.0	1.30E-06
300.0	1.08E-06
350.0	9.18E-07
400.0	7.97E-07
450.0	7.03E-07
500.0	6.28E-07
550.0	5.66E-07
600.0	5.15E-07
650.0	4.71E-07
700.0	4.34E-07
750.0	4.02E-07
800.0	3.73E-07
850.0	3.48E-07
900.0	3.26E-07
950.0	3.06E-07
1000.0	2.89E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	**Causewayside,**			
Site Address:	**Edinburgh**			
Completed by:	**LR**			
Date:	**21.07.23**		**Version:**	**3.2**
Contaminant	**Benzo(k)fluoranthene**			
Target Concentration (C_T)	**0.00000017**	**mg/l**	**Origin of C_T:**	**WAT-SG-53 (April 2020)**

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Benzo(k)fluoranthene		from Level 1
Target Concentration	C_T	1.70E-07	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	2.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	4.27E+03	days	*Howard et al*
Calculated decay rate		1.62E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	3.97E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.10E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	3.97E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

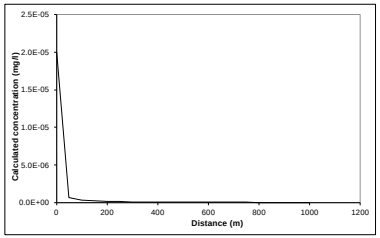
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.92E+02	fraction
Decay rate used		8.44E-07	d^{-1}
Rate of contaminant flow due to retardation	u	1.55E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	3.46E-08	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**5.78E+02**	

Remedial Targets

Remedial Target		**9.83E-05**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**3.46E-08**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	2.0E-05
50.0	7.15E-07
100.0	3.60E-07
150.0	2.40E-07
200.0	1.80E-07
250.0	1.44E-07
300.0	1.19E-07
350.0	1.02E-07
400.0	8.92E-08
450.0	7.91E-08
500.0	7.10E-08
550.0	6.44E-08
600.0	5.89E-08
650.0	5.42E-08
700.0	5.02E-08
750.0	4.67E-08
800.0	4.37E-08
850.0	4.10E-08
900.0	3.86E-08
950.0	3.65E-08
1000.0	3.46E-08

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Chrysene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Chrysene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.10E-04	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	2.00E+03	days	*Howard et al*
Calculated decay rate		3.47E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	3.59E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	9.92E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	3.59E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

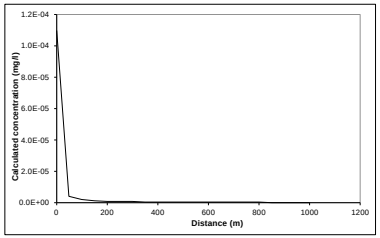
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.74E+02	fraction
Decay rate used		1.99E-06	d^{-1}
Rate of contaminant flow due to retardation	u	1.71E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.79E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**6.15E+02**	

Remedial Targets

Remedial Target		**6.15E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.79E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.1E-04
50.0	3.92E-06
100.0	1.97E-06
150.0	1.31E-06
200.0	9.77E-07
250.0	7.78E-07
300.0	6.45E-07
350.0	5.50E-07
400.0	4.79E-07
450.0	4.23E-07
500.0	3.79E-07
550.0	3.42E-07
600.0	3.12E-07
650.0	2.87E-07
700.0	2.65E-07
750.0	2.46E-07
800.0	2.29E-07
850.0	2.14E-07
900.0	2.01E-07
950.0	1.89E-07
1000.0	1.79E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Copper			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Copper		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.68E-03	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	1.00E+100	days	*Metal - assumed no decay*
Calculated decay rate		6.93E-101	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	6.75E+00	m	*50% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	8.18E+03	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	3.94E+04	fraction
Decay rate used		1.76E-105	d^{-1}
Rate of contaminant flow due to retardation	u	7.55E-05	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.98E-05	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**8.46E+01**	

Remedial Targets

Remedial Target		**8.46E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.98E-05**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

User specified value for partition coefficient

Entry if specify partition coefficient (option)

Soil water partition coefficient — Kd — 8.18E+03 l/kg

Entry for non-polar organic chemicals (option)

Fraction of organic carbon in aquifer — foc — fraction

Organic carbon partition coefficient — Koc — l/kg

Entry for ionic organic chemicals (option)

Sorption coefficient for related species — $K_{oc,n}$ — l/kg

Sorption coefficient for ionised species — $K_{oc,i}$ — l/kg

pH value — pH

acid dissociation constant — pKa

Fraction of organic carbon in aquifer — foc — fraction

Soil water partition coefficient — Kd — 8.18E+03 — l/kg

Define dispersivity (click brown cell and use pull down list)

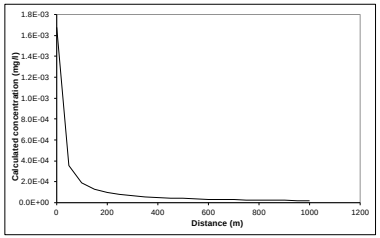
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.7E-03
50.0	3.53E-04
100.0	1.88E-04
150.0	1.28E-04
200.0	9.68E-05
250.0	7.79E-05
300.0	6.52E-05
350.0	5.61E-05
400.0	4.92E-05
450.0	4.38E-05
500.0	3.94E-05
550.0	3.59E-05
600.0	3.29E-05
650.0	3.04E-05
700.0	2.83E-05
750.0	2.64E-05
800.0	2.48E-05
850.0	2.33E-05
900.0	2.20E-05
950.0	2.09E-05
1000.0	1.98E-05

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Fluoranthene			
Target Concentration (C_T)	0.0000063	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Fluoranthene		from Level 1
Target Concentration	C_T	6.30E-06	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	9.80E-05	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	8.80E+02	days	*Howard et al*
Calculated decay rate		7.88E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	2.14E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	5.90E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	2.14E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

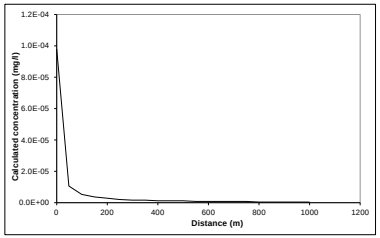
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.04E+02	fraction
Decay rate used		7.59E-06	d^{-1}
Rate of contaminant flow due to retardation	u	2.86E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	4.48E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.19E+02**	

Remedial Targets

Remedial Target		**1.38E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**4.48E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	9.8E-05
50.0	1.07E-05
100.0	5.48E-06
150.0	3.65E-06
200.0	2.72E-06
250.0	2.15E-06
300.0	1.78E-06
350.0	1.51E-06
400.0	1.30E-06
450.0	1.14E-06
500.0	1.02E-06
550.0	9.13E-07
600.0	8.27E-07
650.0	7.54E-07
700.0	6.91E-07
750.0	6.37E-07
800.0	5.90E-07
850.0	5.48E-07
900.0	5.11E-07
950.0	4.78E-07
1000.0	4.48E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Fluorene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Fluorene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	2.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.20E+02	days	*Howard et al*
Calculated decay rate		5.78E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	5.36E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.48E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	5.36E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

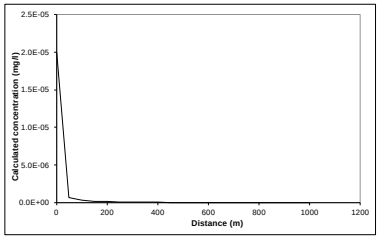
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	2.59E+02	fraction
Decay rate used		2.23E-05	d^{-1}
Rate of contaminant flow due to retardation	u	1.15E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	6.91E-09	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.90E+03**	

Remedial Targets

Remedial Target		**2.90E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**6.91E-09**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	2.0E-05
50.0	6.59E-07
100.0	3.06E-07
150.0	1.89E-07
200.0	1.30E-07
250.0	9.60E-08
300.0	7.37E-08
350.0	5.81E-08
400.0	4.68E-08
450.0	3.83E-08
500.0	3.17E-08
550.0	2.65E-08
600.0	2.24E-08
650.0	1.90E-08
700.0	1.63E-08
750.0	1.40E-08
800.0	1.20E-08
850.0	1.04E-08
900.0	9.07E-09
950.0	7.90E-09
1000.0	6.91E-09

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	**Causewayside,**			
Site Address:	**Edinburgh**			
Completed by:	**LR**			
Date:	**21.07.23**		**Version:**	**3.2**
Contaminant	**Indeno(1,2,3,c-d)pyrene**			
Target Concentration (C_T)	**0.00000017**	**mg/l**	**Origin of C_T:**	**WAT-SG-53 (April 2020)**

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Indeno(1,2,3,c-d)pyrene		from Level 1
Target Concentration	C_T	1.70E-07	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	2.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.46E+03	days	*Howard et al*
Calculated decay rate		4.75E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	2.51E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	6.92E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	2.51E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

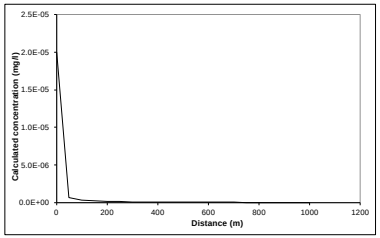
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.22E+02	fraction
Decay rate used		3.90E-06	d^{-1}
Rate of contaminant flow due to retardation	u	2.44E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	3.12E-08	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**6.41E+02**	

Remedial Targets

Remedial Target		**1.09E-04**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**3.12E-08**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	2.0E-05
50.0	7.11E-07
100.0	3.56E-07
150.0	2.36E-07
200.0	1.76E-07
250.0	1.40E-07
300.0	1.16E-07
350.0	9.86E-08
400.0	8.56E-08
450.0	7.55E-08
500.0	6.74E-08
550.0	6.08E-08
600.0	5.53E-08
650.0	5.07E-08
700.0	4.67E-08
750.0	4.33E-08
800.0	4.02E-08
850.0	3.76E-08
900.0	3.52E-08
950.0	3.31E-08
1000.0	3.12E-08

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Phenanthrene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Phenanthrene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.74E-04	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	4.02E+02	days	*Howard et al*
Calculated decay rate		1.73E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.65E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	4.55E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.65E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

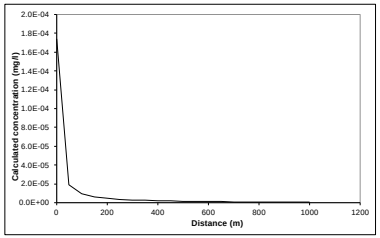
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	8.03E+01	fraction
Decay rate used		2.15E-05	d^{-1}
Rate of contaminant flow due to retardation	u	3.70E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	5.95E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.93E+02**	

Remedial Targets

Remedial Target		**2.93E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**5.95E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.7E-04
50.0	1.88E-05
100.0	9.46E-06
150.0	6.20E-06
200.0	4.55E-06
250.0	3.56E-06
300.0	2.89E-06
350.0	2.41E-06
400.0	2.06E-06
450.0	1.78E-06
500.0	1.56E-06
550.0	1.38E-06
600.0	1.23E-06
650.0	1.11E-06
700.0	1.00E-06
750.0	9.09E-07
800.0	8.29E-07
850.0	7.59E-07
900.0	6.98E-07
950.0	6.43E-07
1000.0	5.95E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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The calculation of equations in this worksheet has been independently checked by Entec (UK) Ltd on behalf of the Environment Agency.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Pyrene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Pyrene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.10E-04	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	3.80E+03	days	*Howard et al*
Calculated decay rate		1.83E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	2.43E+01	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	6.72E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	2.43E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

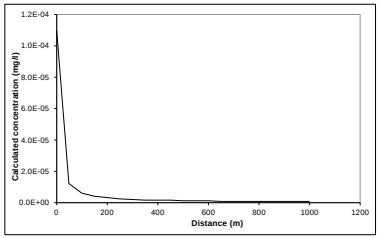
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.18E+02	fraction
Decay rate used		1.55E-06	d^{-1}
Rate of contaminant flow due to retardation	u	2.52E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	6.13E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**1.79E+02**	

Remedial Targets

Remedial Target		**1.79E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**6.13E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.1E-04
50.0	1.22E-05
100.0	6.28E-06
150.0	4.22E-06
200.0	3.17E-06
250.0	2.54E-06
300.0	2.12E-06
350.0	1.81E-06
400.0	1.58E-06
450.0	1.40E-06
500.0	1.26E-06
550.0	1.14E-06
600.0	1.04E-06
650.0	9.62E-07
700.0	8.91E-07
750.0	8.29E-07
800.0	7.75E-07
850.0	7.27E-07
900.0	6.85E-07
950.0	6.47E-07
1000.0	6.13E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Selenium			
Target Concentration (C_T)	0.00025	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
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The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Selenium		from Level 1
Target Concentration	C_T	2.50E-04	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	7.46E-04	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	1.00E+100	days	*Metal - assumed no decay*
Calculated decay rate		6.93E-101	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	6.75E+00	m	*50% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	2.00E+03	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	9.61E+03	fraction
Decay rate used		7.21E-105	d^{-1}
Rate of contaminant flow due to retardation	u	3.09E-04	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	8.81E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**8.46E+01**	

Remedial Targets

Remedial Target		**2.12E-02**	**mg/l**
Domenico - Steady state			
Distance to compliance point		**1000**	**m**
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**8.81E-06**	**mg/l**

For comparison with measured groundwater concentration.

Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

User specified value for partition coefficient

Entry if specify partition coefficient (option)

Soil water partition coefficient — Kd — 2.00E+03 l/kg

Entry for non-polar organic chemicals (option)

Fraction of organic carbon in aquifer — foc — fraction

Organic carbon partition coefficient — Koc — l/kg

Entry for ionic organic chemicals (option)

Sorption coefficient for related species — $K_{OC,n}$ — l/kg

Sorption coefficient for ionised species — $K_{OC,I}$ — l/kg

pH value — pH

acid dissociation constant — pKa

Fraction of organic carbon in aquifer — foc — fraction

Soil water partition coefficient — Kd — 2.00E+03 l/kg

Define dispersivity (click brown cell and use pull down list)

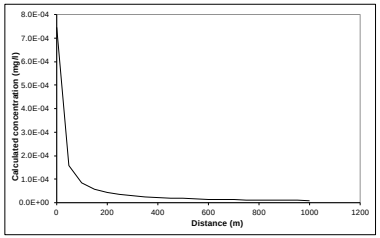
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	7.5E-04
50.0	1.57E-04
100.0	8.33E-05
150.0	5.67E-05
200.0	4.30E-05
250.0	3.46E-05
300.0	2.90E-05
350.0	2.49E-05
400.0	2.18E-05
450.0	1.94E-05
500.0	1.75E-05
550.0	1.59E-05
600.0	1.46E-05
650.0	1.35E-05
700.0	1.26E-05
750.0	1.17E-05
800.0	1.10E-05
850.0	1.04E-05
900.0	9.79E-06
950.0	9.27E-06
1000.0	8.81E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Total Cyanide			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu) | Variable | Value | Unit | Source

Contaminant		Total Cyanide		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.70E-03	mg/l	*Max conc recorded within BH01B/RBH01B*
Half life for degradation of contaminant in water	$t_{1/2}$	1.00E+100	days	*Metal - assumed no decay*
Calculated decay rate		6.93E-101	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH01B/RBH01B*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.00E+03	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

User specified value for partition coefficient

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd	1.00E+03	l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc		fraction
Organic carbon partition coefficient	Koc		l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.00E+03	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

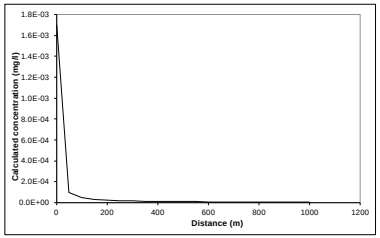
Calculated Parameters | Variable

Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	4.82E+03	fraction
Decay rate used		1.44E-104	d^{-1}
Rate of contaminant flow due to retardation	u	6.17E-04	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	5.05E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.37E+02**	

Remedial Targets

Remedial Target		**3.37E-01**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**5.05E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.7E-03
50.0	9.79E-05
100.0	4.97E-05
150.0	3.33E-05
200.0	2.51E-05
250.0	2.01E-05
300.0	1.68E-05
350.0	1.44E-05
400.0	1.26E-05
450.0	1.12E-05
500.0	1.01E-05
550.0	9.16E-06
600.0	8.40E-06
650.0	7.76E-06
700.0	7.20E-06
750.0	6.72E-06
800.0	6.30E-06
850.0	5.93E-06
900.0	5.60E-06
950.0	5.31E-06
1000.0	5.05E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Zinc			
Target Concentration (C_T)	0.0109	mg/l	Origin of C_T:	WAT-SG-53 (April 2020)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
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The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu) | Variable | Value | Unit | Source

Contaminant		Zinc		from Level 1
Target Concentration	C_T	1.09E-02	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	3.92E-02	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	1.00E+100	days	*Metal - assumed no decay*
Calculated decay rate		6.93E-101	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	6.75E+00	m	*50% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	1.00E+03	m	*Distance from site to the Pow Burn/Braid*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	3.14E+03	l/kg	*see options*
Longitudinal dispersivity	ax	1.00E+02	m	*see options*
Transverse dispersivity	az	1.00E+01	m	*see options*
Vertical dispersivity	ay	1.00E+00	m	*see options*

Calculated Parameters | Variable

Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.51E+04	fraction
Decay rate used		4.59E-105	d^{-1}
Rate of contaminant flow due to retardation	u	1.97E-04	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	4.63E-04	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**8.46E+01**	

Remedial Targets

Remedial Target		**9.23E-01**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**1000**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**4.63E-04**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

User specified value for partition coefficient

Entry if specify partition coefficient (option)

Soil water partition coefficient — Kd — 3.14E+03 l/kg

Entry for non-polar organic chemicals (option)

Fraction of organic carbon in aquifer — foc — fraction

Organic carbon partition coefficient — Koc — l/kg

Entry for ionic organic chemicals (option)

Sorption coefficient for related species — $K_{oc,n}$ — l/kg

Sorption coefficient for ionised species — $K_{oc,i}$ — l/kg

pH value — pH

acid dissociation constant — pKa

Fraction of organic carbon in aquifer — foc — fraction

Soil water partition coefficient — Kd — 3.14E+03 l/kg

Define dispersivity (click brown cell and use pull down list)

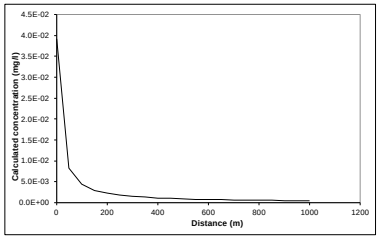
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	1.00E+02	1.18E+01	m
Transverse dispersivity	**az**	0.00E+00	1.00E+01	1.18E+00	m
Vertical dispersivity	**ay**	0.00E+00	1.00E+00	1.18E-01	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	3.9E-02
50.0	8.24E-03
100.0	4.38E-03
150.0	2.98E-03
200.0	2.26E-03
250.0	1.82E-03
300.0	1.52E-03
350.0	1.31E-03
400.0	1.15E-03
450.0	1.02E-03
500.0	9.20E-04
550.0	8.38E-04
600.0	7.69E-04
650.0	7.10E-04
700.0	6.60E-04
750.0	6.16E-04
800.0	5.78E-04
850.0	5.44E-04
900.0	5.14E-04
950.0	4.87E-04
1000.0	4.63E-04

Groundwater Assessment

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

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Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C10-C12			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
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R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Aliphatic C10-C12		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	5.70E-03	mg/l	*Max conc recorded within BH01B/RBH01B*
Half life for degradation of contaminant in water	$t_{1/2}$	1.02E+02	days	*Howard et al*
Calculated decay rate		6.80E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH01B/RBH01B*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.14E+04	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	3.16E+05	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.14E+04	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

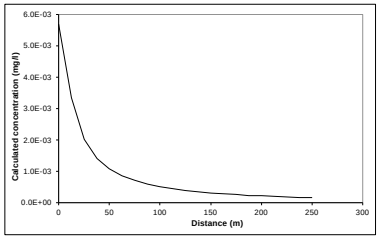
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	5.51E+04	fraction
Decay rate used		1.23E-07	d^{-1}
Rate of contaminant flow due to retardation	u	5.39E-05	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.54E-04	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.71E+01**	

Remedial Targets

Remedial Target		**3.71E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.54E-04**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	5.7E-03
12.5	3.34E-03
25.0	2.02E-03
37.5	1.41E-03
50.0	1.07E-03
62.5	8.57E-04
75.0	7.07E-04
87.5	5.97E-04
100.0	5.13E-04
112.5	4.46E-04
125.0	3.93E-04
137.5	3.50E-04
150.0	3.13E-04
162.5	2.82E-04
175.0	2.56E-04
187.5	2.33E-04
200.0	2.13E-04
212.5	1.95E-04
225.0	1.80E-04
237.5	1.66E-04
250.0	1.54E-04

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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The calculation of equations in this worksheet has been independently checked by Entec (UK) Ltd on behalf of the Environment Agency.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C12-C16			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Aliphatic C12-C16		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	6.98E-03	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	4.40E+02	days	*Howard et al*
Calculated decay rate		1.58E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.81E+05	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	5.01E+06	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.81E+05	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

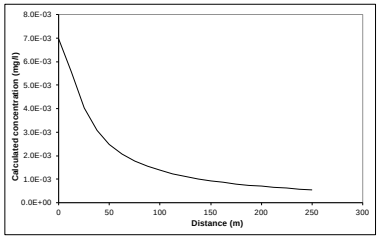
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	8.74E+05	fraction
Decay rate used		1.80E-09	d^{-1}
Rate of contaminant flow due to retardation	u	3.40E-06	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	5.51E-04	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**1.27E+01**	

Remedial Targets

Remedial Target		**1.27E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**5.51E-04**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	7.0E-03
12.5	5.55E-03
25.0	4.01E-03
37.5	3.08E-03
50.0	2.49E-03
62.5	2.08E-03
75.0	1.78E-03
87.5	1.55E-03
100.0	1.38E-03
112.5	1.23E-03
125.0	1.12E-03
137.5	1.02E-03
150.0	9.34E-04
162.5	8.62E-04
175.0	8.00E-04
187.5	7.46E-04
200.0	6.98E-04
212.5	6.55E-04
225.0	6.17E-04
237.5	5.82E-04
250.0	5.51E-04

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C16-C21			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Aliphatic C16-C21		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.56E-02	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	5.30E+02	days	*Howard et al*
Calculated decay rate		1.31E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.44E+07	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	3.98E+08	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.44E+07	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

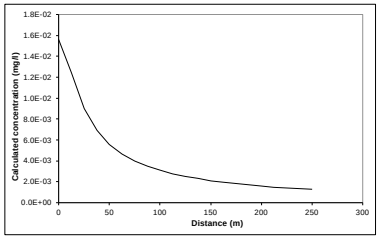
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	6.94E+07	fraction
Decay rate used		1.88E-11	d^{-1}
Rate of contaminant flow due to retardation	u	4.28E-08	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.26E-03	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**1.24E+01**	

Remedial Targets

Remedial Target		**1.24E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.26E-03**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.6E-02
12.5	1.24E-02
25.0	8.99E-03
37.5	6.92E-03
50.0	5.59E-03
62.5	4.67E-03
75.0	4.00E-03
87.5	3.50E-03
100.0	3.10E-03
112.5	2.78E-03
125.0	2.52E-03
137.5	2.30E-03
150.0	2.11E-03
162.5	1.95E-03
175.0	1.82E-03
187.5	1.69E-03
200.0	1.59E-03
212.5	1.49E-03
225.0	1.41E-03
237.5	1.33E-03
250.0	1.26E-03

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aliphatic C21-C35			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Aliphatic C21-C35		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.27E-01	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	7.30E+02	days	*Howard et al*
Calculated decay rate		9.50E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.50E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.44E+07	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	6.94E+07	fraction
Decay rate used		1.37E-11	d^{-1}
Rate of contaminant flow due to retardation	u	4.28E-08	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.05E-02	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**1.20E+01**	

Remedial Targets

Remedial Target		**1.20E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.05E-02**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	3.98E+08	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.44E+07	l/kg

Define dispersivity (click brown cell and use pull down list)

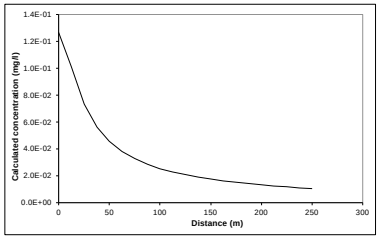
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.3E-01
12.5	1.01E-01
25.0	7.32E-02
37.5	5.64E-02
50.0	4.56E-02
62.5	3.82E-02
75.0	3.28E-02
87.5	2.87E-02
100.0	2.55E-02
112.5	2.29E-02
125.0	2.08E-02
137.5	1.90E-02
150.0	1.75E-02
162.5	1.62E-02
175.0	1.50E-02
187.5	1.41E-02
200.0	1.32E-02
212.5	1.24E-02
225.0	1.17E-02
237.5	1.11E-02
250.0	1.05E-02

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Aromatic C21-C35			
Target Concentration (C_T)	0.001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Aromatic C21-C35		from Level 1
Target Concentration	C_T	1.00E-03	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	5.90E-02	mg/l	*Max conc recorded within BH01B/RBH01B*
Half life for degradation of contaminant in water	$t_{1/2}$	7.30E+02	days	*Howard et al*
Calculated decay rate		9.50E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH01B/RBH01B*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	4.56E+03	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.26E+05	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	4.56E+03	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

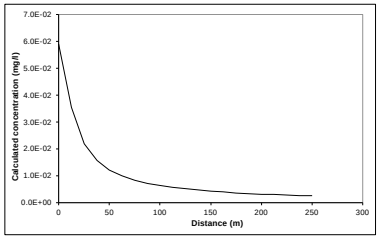
Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	2.20E+04	fraction
Decay rate used		4.32E-08	d^{-1}
Rate of contaminant flow due to retardation	u	1.35E-04	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	2.53E-03	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**2.33E+01**	

Remedial Targets

Remedial Target		**2.33E-02**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**2.53E-03**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	5.9E-02
12.5	3.54E-02
25.0	2.19E-02
37.5	1.57E-02
50.0	1.22E-02
62.5	9.96E-03
75.0	8.41E-03
87.5	7.26E-03
100.0	6.38E-03
112.5	5.69E-03
125.0	5.13E-03
137.5	4.67E-03
150.0	4.28E-03
162.5	3.94E-03
175.0	3.66E-03
187.5	3.41E-03
200.0	3.19E-03
212.5	3.00E-03
225.0	2.82E-03
237.5	2.67E-03
250.0	2.53E-03

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	**Causewayside,**			
Site Address:	**Edinburgh**			
Completed by:	**LR**			
Date:	**21.07.23**		**Version:**	**3.2**
Contaminant	**Benzo(a)anthracene**			
Target Concentration (C_T)	**0.00001**	**mg/l**	**Origin of C_T:**	**Laboratory Limit of Detection (LoD)**

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)

	Variable	Value	Unit	Source
Contaminant		Benzo(a)anthracene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	4.40E-05	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	1.36E+03	days	*Howard et al*
Calculated decay rate		5.09E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.60E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.35E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	5.36E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Calculated Parameters

	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	2.59E+02	fraction
Decay rate used		1.96E-06	d^{-1}
Rate of contaminant flow due to retardation	u	1.15E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	3.89E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**1.13E+01**	

Remedial Targets

Remedial Target		**1.13E-04**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**3.89E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.48E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	5.36E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

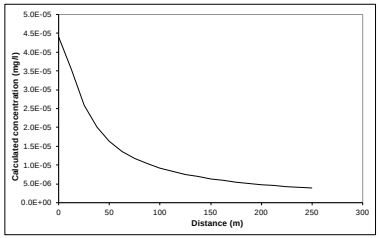
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	4.4E-05
12.5	3.53E-05
25.0	2.58E-05
37.5	2.00E-05
50.0	1.63E-05
62.5	1.37E-05
75.0	1.18E-05
87.5	1.03E-05
100.0	9.19E-06
112.5	8.28E-06
125.0	7.52E-06
137.5	6.89E-06
150.0	6.36E-06
162.5	5.90E-06
175.0	5.50E-06
187.5	5.15E-06
200.0	4.84E-06
212.5	4.57E-06
225.0	4.32E-06
237.5	4.10E-06
250.0	3.89E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Benzo(a)pyrene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	WAT-PS-10-01 (August 2014)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Benzo(a)pyrene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	4.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.06E+03	days	*Howard et al*
Calculated decay rate		6.55E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.46E+02	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	4.02E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.46E+02	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

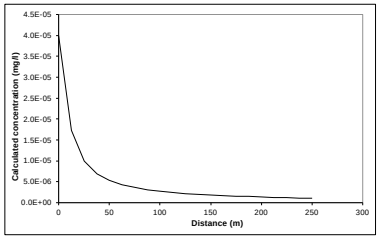
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	7.03E+02	fraction
Decay rate used		9.32E-07	d^{-1}
Rate of contaminant flow due to retardation	u	4.23E-03	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	1.09E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.67E+01**	

Remedial Targets

Remedial Target		**3.67E-04**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**1.09E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	4.0E-05
12.5	1.73E-05
25.0	9.97E-06
37.5	6.97E-06
50.0	5.35E-06
62.5	4.33E-06
75.0	3.64E-06
87.5	3.13E-06
100.0	2.75E-06
112.5	2.45E-06
125.0	2.21E-06
137.5	2.00E-06
150.0	1.84E-06
162.5	1.69E-06
175.0	1.57E-06
187.5	1.47E-06
200.0	1.37E-06
212.5	1.29E-06
225.0	1.22E-06
237.5	1.15E-06
250.0	1.09E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Benzo(b)fluoranthene			
Target Concentration (C_T)	0.0001	mg/l	Origin of C_T:	WAT-PS-10-01 (August 2014)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Benzo(b)fluoranthene		from Level 1
Target Concentration	C_T	1.00E-04	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.00E-04	mg/l	*Max concentration recorded in BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.22E+03	days	*Howard et al*
Calculated decay rate		5.69E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.60E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	4.41E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.60E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

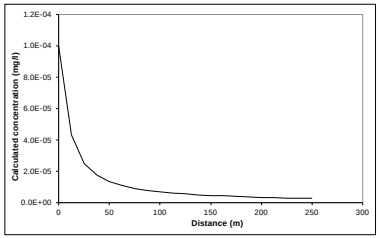
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	7.80E+01	fraction
Decay rate used		7.29E-06	d^{-1}
Rate of contaminant flow due to retardation	u	3.81E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	2.74E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.64E+01**	

Remedial Targets

Remedial Target		**3.64E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**2.74E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.0E-04
12.5	4.33E-05
25.0	2.49E-05
37.5	1.74E-05
50.0	1.34E-05
62.5	1.09E-05
75.0	9.12E-06
87.5	7.85E-06
100.0	6.89E-06
112.5	6.14E-06
125.0	5.53E-06
137.5	5.03E-06
150.0	4.61E-06
162.5	4.26E-06
175.0	3.95E-06
187.5	3.68E-06
200.0	3.45E-06
212.5	3.24E-06
225.0	3.06E-06
237.5	2.89E-06
250.0	2.74E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	**Causewayside,**			
Site Address:	**Edinburgh**			
Completed by:	**LR**			
Date:	**21.07.23**		**Version:**	**3.2**
Contaminant	**Benzo(ghi)perylene**			
Target Concentration (C_T)	**0.0001**	**mg/l**	**Origin of C_T:**	**WAT-PS-10-01 (August 2014)**

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu) | Variable | Value | Unit | Source

Contaminant		Benzo(ghi)perylene		from Level 1
Target Concentration	C_T	1.00E-04	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	1.20E-04	mg/l	*Max concentration encountered (BH02A/RBH02A)*
Half life for degradation of contaminant in water	$t_{1/2}$	1.31E+03	days	*Howard et al*
Calculated decay rate		5.28E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	1.89E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Calculated Parameters | Variable

Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	9.22E+01	fraction
Decay rate used		5.72E-06	d^{-1}
Rate of contaminant flow due to retardation	u	3.22E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	3.30E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.63E+01**	

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	5.23E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	1.89E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

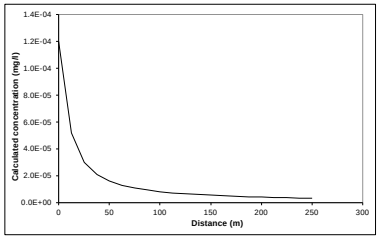
Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	1.2E-04
12.5	5.20E-05
25.0	2.99E-05
37.5	2.09E-05
50.0	1.61E-05
62.5	1.30E-05
75.0	1.10E-05
87.5	9.44E-06
100.0	8.28E-06
112.5	7.38E-06
125.0	6.65E-06
137.5	6.05E-06
150.0	5.55E-06
162.5	5.12E-06
175.0	4.75E-06
187.5	4.43E-06
200.0	4.15E-06
212.5	3.90E-06
225.0	3.68E-06
237.5	3.48E-06
250.0	3.30E-06

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Remedial Targets

Remedial Target		**3.63E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**3.30E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Chrysene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
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It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Chrysene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	5.20E-04	mg/l	*Average concentration from across site*
Half life for degradation of contaminant in water	$t_{1/2}$	2.00E+03	days	*Howard et al*
Calculated decay rate		3.47E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	3.60E+01	m	*Width of site*
Plume thickness at source	Sy	3.38E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	1.30E+01	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	3.59E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	9.92E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	3.59E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

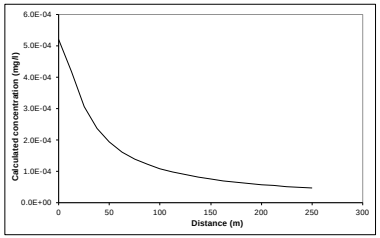
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.74E+02	fraction
Decay rate used		1.99E-06	d^{-1}
Rate of contaminant flow due to retardation	u	1.71E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	4.67E-05	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**1.11E+01**	

Remedial Targets

Remedial Target		**1.11E-04**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**4.67E-05**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	5.2E-04
12.5	4.18E-04
25.0	3.06E-04
37.5	2.37E-04
50.0	1.93E-04
62.5	1.62E-04
75.0	1.40E-04
87.5	1.23E-04
100.0	1.09E-04
112.5	9.84E-05
125.0	8.95E-05
137.5	8.21E-05
150.0	7.58E-05
162.5	7.03E-05
175.0	6.56E-05
187.5	6.15E-05
200.0	5.78E-05
212.5	5.46E-05
225.0	5.17E-05
237.5	4.90E-05
250.0	4.67E-05

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Fluoranthene			
Target Concentration (C_T)	0.0001	mg/l	Origin of C_T:	WAT-PS-10-01 (August 2014)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Fluoranthene		from Level 1
Target Concentration	C_T	1.00E-04	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	3.00E-04	mg/l	*Max concentration recorded in BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	8.80E+02	days	*Howard et al*
Calculated decay rate		7.88E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	2.14E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	5.90E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	2.14E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

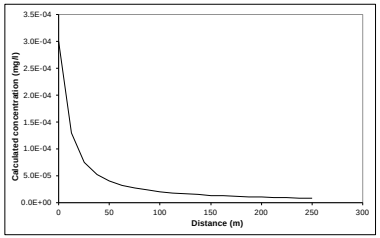
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.04E+02	fraction
Decay rate used		7.59E-06	d^{-1}
Rate of contaminant flow due to retardation	u	2.86E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	8.08E-06	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.71E+01**	

Remedial Targets

Remedial Target		**3.71E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**8.08E-06**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	3.0E-04
12.5	1.30E-04
25.0	7.47E-05
37.5	5.22E-05
50.0	4.00E-05
62.5	3.24E-05
75.0	2.72E-05
87.5	2.34E-05
100.0	2.05E-05
112.5	1.83E-05
125.0	1.64E-05
137.5	1.49E-05
150.0	1.37E-05
162.5	1.26E-05
175.0	1.17E-05
187.5	1.09E-05
200.0	1.02E-05
212.5	9.58E-06
225.0	9.03E-06
237.5	8.53E-06
250.0	8.08E-06

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Fluorene			
Target Concentration (C_T)	0.00001	mg/l	Origin of C_T:	Laboratory Limit of Detection (LoD)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Fluorene		from Level 1
Target Concentration	C_T	1.00E-05	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	2.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.20E+02	days	*Howard et al*
Calculated decay rate		5.78E-03	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	5.36E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	1.48E+03	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	5.36E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

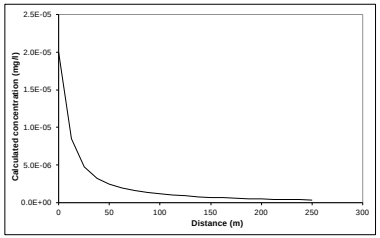
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	2.59E+02	fraction
Decay rate used		2.23E-05	d^{-1}
Rate of contaminant flow due to retardation	u	1.15E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	3.62E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**5.53E+01**	

Remedial Targets

Remedial Target		**5.53E-04**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**3.62E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	2.0E-05
12.5	8.48E-06
25.0	4.79E-06
37.5	3.28E-06
50.0	2.46E-06
62.5	1.96E-06
75.0	1.61E-06
87.5	1.36E-06
100.0	1.17E-06
112.5	1.02E-06
125.0	8.99E-07
137.5	8.00E-07
150.0	7.19E-07
162.5	6.49E-07
175.0	5.90E-07
187.5	5.39E-07
200.0	4.94E-07
212.5	4.55E-07
225.0	4.20E-07
237.5	3.90E-07
250.0	3.62E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	Causewayside,			
Site Address:	Edinburgh			
Completed by:	LR			
Date:	21.07.23		Version:	3.2
Contaminant	Indeno(1,2,3,c-d)pyrene			
Target Concentration (C_T)	0.0001	mg/l	Origin of C_T:	WAT-PS-10-01 (August 2014)

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.

R&D Publication 20 Remedial Targets Worksheet, Release 3.2

Level 3 - Groundwater

See Note

Input Parameters (using pull down menu)	Variable	Value	Unit	Source
Contaminant		Indeno(1,2,3,c-d)pyrene		from Level 1
Target Concentration	C_T	1.00E-04	mg/l	from Level 1

Select analytical solution (click on brown cell below, then on pull-down menu)

Domenico - Steady state — Equations in HRA publication

Approach for simulating vertical dispersion: Simulate vertical dispersion in 1 direction

Select nature of decay rate (click on brown cell below, then on pull-down menu)

Approach for simulating degradation of pollutants: Apply degradation rate to dissolved pollutants only

				Source of parameter value
Initial contaminant concentration in groundwater at plume core	C_0	2.00E-05	mg/l	*Max conc recorded within BH02A/RBH02A*
Half life for degradation of contaminant in water	$t_{1/2}$	1.46E+03	days	*Howard et al*
Calculated decay rate		4.75E-04	$days^{-1}$	
Width of plume in aquifer at source (perpendicular to flow)	Sz	1.75E+01	m	*Area of influence around BH02A/RBH02A*
Plume thickness at source	Sy	2.08E+00	m	*25% saturated aquifer thickness*
Saturated aquifer thickness	da	8.30E+00	m	*Max groundwater thickness encountered*
Bulk density of aquifer materials		2.51E+00	g/cm^3	*structx.com*
Effective porosity of aquifer	n	5.20E-01	fraction	*geotechdata.info*
Hydraulic gradient	i	1.79E-02	fraction	*Calculated between BH01B/RBH01B and BH02A/RBH02A*
Hydraulic conductivity of aquifer	K	8.64E+01	m/d	*Freeze and Cherry (1979)*
Distance to compliance point	x	2.50E+02	m	*WAT-PS-10-01 (August 2014)*
Distance (lateral) to compliance point perpendicular to flow direction	z		m	
Distance (depth) to compliance point perpendicular to flow direction	y		m	
Time since pollutant entered groundwater	t	1.00E+100	days	*time variant options only*
Parameters values determined from options				
Partition coefficient	Kd	2.51E+01	l/kg	*see options*
Longitudinal dispersivity	ax	2.50E+01	m	*see options*
Transverse dispersivity	az	2.50E+00	m	*see options*
Vertical dispersivity	ay	2.50E-01	m	*see options*

Select Method for deriving Partition Co-efficient (using pull down menu)

Calculate for non-polar organic chemicals

Entry if specify partition coefficient (option)			
Soil water partition coefficient	Kd		l/kg
Entry for non-polar organic chemicals (option)			
Fraction of organic carbon in aquifer	foc	3.62E-02	fraction
Organic carbon partition coefficient	Koc	6.92E+02	l/kg
Entry for ionic organic chemicals (option)			
Sorption coefficient for related species	$K_{oc,n}$		l/kg
Sorption coefficient for ionised species	$K_{oc,i}$		l/kg
pH value	pH		
acid dissociation constant	pKa		
Fraction of organic carbon in aquifer	foc		fraction
Soil water partition coefficient	Kd	2.51E+01	l/kg

Define dispersivity (click brown cell and use pull down list)

Dispersivities 10%, 1%, 0.1% of pathway length

		Enter value	Calc value	Xu & Eckstein	
Longitudinal dispersivity	**ax**	0.00E+00	2.50E+01	6.86E+00	m
Transverse dispersivity	**az**	0.00E+00	2.50E+00	6.86E-01	m
Vertical dispersivity	**ay**	0.00E+00	2.50E-01	6.86E-02	

Note values of dispersivity must be > 0

For calculated value, assumes ax = 0.1 *x, az = 0.01 * x, ay = 0.001 * x

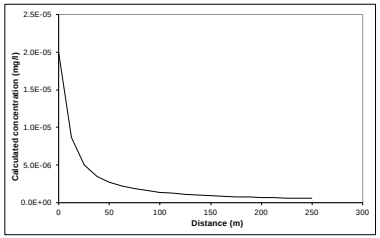
Xu & Eckstein (1995) report ax = $0.83(\log_{10}x)^{2.414}$; az = ax/10, ay = ax/100 are assumed

Calculated Parameters	Variable		
Groundwater flow velocity	v	2.97E+00	m/d
Retardation factor	Rf	1.22E+02	fraction
Decay rate used		3.90E-06	d^{-1}
Rate of contaminant flow due to retardation	u	2.44E-02	m/d
Contaminant concentration at distance x, assuming one-way vertical dispersion	C_{ED}	5.53E-07	mg/l
Attenuation factor (one way vertical dispersion, CO/CED)	**AF**	**3.62E+01**	

Remedial Targets

Remedial Target		**3.62E-03**	**mg/l**	For comparison with measured groundwater concentration.
Domenico - Steady state				
Distance to compliance point		**250**	**m**	
Concentration of contaminant at compliance point	**C_{ED}/C_0**	**5.53E-07**	**mg/l**	Domenico - Steady state

Care should be used when calculating remedial targets using the time variant options as this may result in an overestimate of the remedial target.
The recommended value for time when calculating the remedial target is 9.9E+99.



Note graph assumes plume disperses vertically in one direction only. An alternative solution assuming the centre of the plume is located at the mid-depth of the aquifer is presented in the calculation sheets.

Note

This sheet calculates the Level 3 remedial target for groundwater, based on the distance to the receptor or compliance located down hydraulic gradient of the source Three solution methods are included, the preferred option is Ogata Banks.

By setting a long travel time it will give the steady state solution, which should be used to calculate remedial targets.

The measured groundwater concentration should be compared with the Level 3 remedial target to determine the need for further action.
Note if contaminant is not subject to first order degradation, then set half life as 9.0E+99.

This worksheet should be used if pollutant transport and degradation is best described by a first order reaction. If degradation is best desribed by an electron limited degradation such as oxidation by O2, NO3, SO4 etc than an alternative solution should be used

Site being assessed:	Causewayside,
Completed by:	LR
Date:	21.07.23
Version:	3.2

Calculated concentrations for distance-concentration graph

Domenico - Steady state
From calculation sheet

Distance	Concentration
	mg/l
0	2.0E-05
12.5	8.67E-06
25.0	4.99E-06
37.5	3.49E-06
50.0	2.68E-06
62.5	2.17E-06
75.0	1.83E-06
87.5	1.58E-06
100.0	1.38E-06
112.5	1.23E-06
125.0	1.11E-06
137.5	1.01E-06
150.0	9.27E-07
162.5	8.56E-07
175.0	7.94E-07
187.5	7.41E-07
200.0	6.94E-07
212.5	6.53E-07
225.0	6.16E-07
237.5	5.83E-07
250.0	5.53E-07

Hydrogeological risk assessment for land contamination

Remedial Targets Worksheet , Release 3.2

First released: 2006. Version 3.2: January 2013

This worksheet has been produced in combination with the document 'Remedial Targets Methodology: Hydrogeological risk assessment for land contamination (Environment Agency 2006).

Users of this worksheet should always refer to the User Manual to the Remedial Targets Methodology and to relevant guidance on UK legislation and policy, in order to understand how this procedure should be applied in an appropriate context.

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The calculation of equations in this worksheet has been independently checked by Entec (UK) Ltd on behalf of the Environment Agency.

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IMPORTANT: To enable MS Excel worksheet, click Tools, Add -Ins, Analysis Tool Pak and Analysis Tool Pak-VBA (to calculate error functions

Details to be completed for each assessment				
Site Name:	**Causewayside,**			
Site Address:	**Edinburgh**			
Completed by:	**LR**			
Date:	**21.07.23**		**Version:**	**3.2**
Contaminant	**Naphthalene**			
Target Concentration (C_T)	**0.00005**	**mg/l**	**Origin of C_T:**	**Laboratory Limit of Detection (LoD)**

This worksheet can be used to determine remedial targets for soils (Worksheets Level 1 Soil, Level 2 and Level 3 Soil) or to determine remedial targets for groundwater (Level 3 Groundwater). For Level 3, parameter values must be entered separately dependent on whether the assessment is for soil or groundwater. For soil, remedial targets are calculated as either mg/kg (for comparision with soil measurements) or mg/l (for comparison with leaching tests or pore water concentrations).
Site details entered on this page are automatically copied to Level 1, 2 and 3 Worksheets.
Worksheet options are identified by brown background and employ a pull-down menus. Data entry are identified as blue background.
Data origin / justification should be noted in cells coloured yellow and fully documented in subsequent reports.
Data carried forward from an earlier worksheet are identified by a light green background
It is recommended that a copy of the original worksheet is saved (all data fields in the original copy are blank).

The spreadsheet also includes a porosity calculation worksheet, a soil impact calculation worksheet and a worksheet that performs some simple hydrogeological calculations.